

Prevention and Management of Acute Kidney Injury

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CHAPTER OUTLINE

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DEFINITION OF ACUTE KIDNEY INJURY

Acute kidney injury (AKI) is a heterogeneous syndrome defined by rapid (hours to days) decline in the glomerular filtration rate (GFR), resulting in the retention of metabolic waste products, including urea and creatinine, and dysregulation of fluid, electrolyte, and acid-base homeostasis.¹ Although often considered a discrete syndrome, AKI represents a broad constellation of pathophysiologic processes of varied severity and cause. These include decreases in the GFR as the result of hemodynamic perturbations that disrupt normal renal perfusion without causing parenchymal injury, partial or complete obstruction to urinary flow, and a range of processes with characteristic patterns of glomerular, interstitial, tubular, or vascular parenchymal injury. AKI occurs in a heterogeneous patient population—genetics, age, kidney functional status, accompanying comorbidities—and the cause is often multifactorial.

The term *acute kidney injury* has largely supplanted the older terminology of acute renal failure (ARF). This change reflects recognition of serious shortcomings with the older terminology. Acute renal failure suggested a dichotomous relationship between normal kidney function and overt organ failure; in contrast, AKI captures the growing body of data associating small to moderate acute and transient decrements in kidney function with serious, adverse outcomes. Although the AKI terminology does emphasize the graded aspect of acute kidney disease, it should be recognized that this terminology is also imperfect. The term *injury* can be construed to imply the presence of parenchymal organ damage, which

may be absent in a variety of settings associated with an acute decline in kidney function, such as early obstructive disease and prerenal azotemia associated with volume depletion. Although the term *acute kidney dysfunction* might better characterize the entire spectrum of the syndrome, AKI is the term that has been adopted by consensus and is now increasingly used in the medical literature.^{2,3} In this chapter, AKI will be used to describe the entire spectrum of the syndrome. Although, in clinical practice, the term *acute tubular necrosis* (ATN) is often used synonymously with AKI, these terms should not be used interchangeably. Although ATN is the most common form of intrinsic AKI, particularly in critically ill patients, it represents only one of multiple forms of AKI. In addition, there may be a lack of concordance between the clinical syndrome and the classic histopathologic findings of ATN.^{4,5}

Decreased urine output is a cardinal (although not universal) manifestation of AKI, and patients are often classified based on urine flow rates as nonoliguric (urine output >400 mL/day), oliguric (urine output <400 mL/day), or anuric (urine output <100 mL/day).⁶ Transient oliguria may occur in the absence of significant decrements in kidney function, because increased tubular salt and water reabsorption is a normal physiologic response to volume depletion. In contradistinction, persistent oliguria, despite the presence of adequate intravascular volume, is virtually always a manifestation of AKI, with lower urine volume typically associated with more severe initial renal injury. The categorization of AKI based on urine volume has clinical implications for the development of volume overload, severity of electrolyte disturbances, and overall prognosis. Although oliguric AKI

is associated with a higher mortality risk than nonoliguric AKI, therapeutic interventions to augment urine output (see later) have not been shown to improve patient outcomes.⁷

AKI can develop *de novo* in the setting of intact kidney function or can be superimposed on underlying chronic kidney disease (acute on chronic kidney injury), and the presence of underlying impaired kidney function has been shown to be one of the most important risk factors for the development of AKI.^{8,9} Multiple mechanisms may contribute to this increased susceptibility, including diminished renal functional reserve, impaired salt and water conservation predisposing to intravascular volume contraction, decreased activity of detoxification mechanisms, increasing susceptibility to cytotoxic injury, impaired clearance of potential nephrotoxins, increasing the risk for and/or duration of exposure, and associated macrovascular and microvascular disease, increasing the risk of ischemic injury.

The causes of AKI are usually divided into three broad pathophysiologic categories:

1. Prerenal AKI—diseases characterized by effective hypoperfusion of the kidneys in which there is no parenchymal damage to the kidney (Box 29.1);
2. Intrinsic AKI—diseases involving the renal parenchyma (Table 29.1); and
3. Postrenal (obstructive) AKI—diseases associated with acute obstruction of the urinary tract (Box 29.2)

Box 29.1 Causes of Prerenal Acute Kidney Injury

Intravascular Volume Depletion
 Hemorrhage—trauma, surgery, postpartum, gastrointestinal
 Gastrointestinal losses—diarrhea, vomiting, nasogastric loss
 Renal losses—diuretics, osmotic diuresis, diabetes insipidus,
 Skin and mucous membrane losses—burns, hyperthermia
 Nephrotic syndrome
 Cirrhosis
 Capillary leak
Reduced Cardiac Output
 Cardiogenic shock
 Pericardial diseases—restrictive, constrictive, tamponade
 Congestive heart failure
 Valvular diseases
 Pulmonary diseases—pulmonary hypertension, pulmonary embolism
 Sepsis
Systemic Vasodilation
 Sepsis
 Cirrhosis
 Anaphylaxis
 Medications
Renal Vasoconstriction
 Early sepsis
 Hepatorenal syndrome
 Acute hypercalcemia
 Drugs—norepinephrine, vasopressin, nonsteroidal antiinflammatory drugs (NSAIDs), angiotension-converting enzyme
 Calcineurin inhibitors
 Iodinated Contrast Agents
 Increased Intraabdominal Pressure
 Abdominal Compartment Syndrome

Although these categories are useful for didactic purposes and help inform the initial clinical assessment of patients presenting with AKI, there is often a degree of overlap between these categories. For example, renal hypoperfusion may cause a spectrum of renal injury ranging from prerenal azotemia to overt ATN, depending on its severity and duration. As a result, precise categorization of the cause of AKI in these three groups may not always be possible, and individual patients may transition between categories regarding causation.

The prior absence of a uniform operational definition of AKI impeded epidemiologic studies and hampered clinical evaluations of preventive and therapeutic interventions. Older literature was characterized by multiple definitions using varying absolute and/or relative changes in the serum creatinine concentration, with or without associated decrements in urine output, which made it difficult to compare findings

Table 29.1 Major Causes of Intrinsic Acute Kidney Injury

Type of Injury	Examples
Tubular injury	
• Ischemia due to hypoperfusion	Hypovolemia, sepsis, hemorrhage, cirrhosis, CHF (see Box 29.1)
• Endogenous toxins	Myoglobin, hemoglobin, paraproteinemia, uric acid (see Table 29.3)
• Exogenous toxins	Antibiotics, chemotherapy agents, radiocontrast, phosphate preps. (see Table 29.3)
Tubulointerstitial injury	
• Acute allergic interstitial nephritis	Nonsteroidal antiinflammatory drugs (NSAIDs), antibiotics
• Infections	Viral, bacterial, and fungal infections
• Infiltration	Lymphoma, leukemia, sarcoid
• Allograft rejection	
Glomerular injury	
• Inflammation	Anti-GBM disease, ANCA-associated GN, postinfectious, cryoglobulinemia, membranoproliferative glomerulonephritis, IgA nephropathy, SLE, Henoch-Schonlein purpura, polyarteritis nodosa
• Hematologic	Hemolytic-uremic syndrome, thrombotic thrombocytopenic purpura, medications
Renal microvasculature	Malignant hypertension, toxemia of pregnancy, hypercalcemia, radiocontrast, scleroderma, medications
Large vessels	
• Arterial	Thrombosis, vasculitis, dissection, thromboembolism, atheroembolism, trauma
• Venous	Thrombosis, compression, trauma

ANCA, Antineutrophil cytoplasmic antibody; CHF, congestive heart failure; GBM, glomerular basement membrane; GN, glomerulonephritis; IgA, immunoglobulin A; SLE, systemic lupus erythematosus.

Box 29.2 Causes of Postrenal Acute Kidney Injury

Upper Urinary Tract Extrinsic Causes

Retroperitoneal space—lymph nodes, tumors
 Pelvic or intraabdominal tumors—cervix, uterus, ovary, prostate
 Fibrosis—radiation, drugs, inflammatory
 Ureteral ligation or surgical trauma
 Granulomatous diseases
 Hematoma

Lower Urinary Tract Causes

Prostate—benign prostatic hypertrophy, carcinoma, infection
 Bladder—neck obstruction, calculi, carcinoma, infection (schistosomiasis)
 Functional—neurogenic bladder secondary to spinal cord injury, diabetes, multiple sclerosis, stroke, pharmacologic side effects of drugs (anticholinergics, antidepressants)
 Urethral—posterior urethral valves, strictures, trauma, infections, tuberculosis, tumors

Upper Urinary Tract: Intrinsic Causes

Nephrolithiasis
 Strictures
 Edema
 Debris, blood clots, sloughed papillae, fungal ball
 Malignancy

across studies. In 2002, the Acute Dialysis Quality Improvement Initiative (ADQI) proposed the first consensus definition of AKI. The ADQI work group proposed a classification scheme with three strata based on the magnitude of increases in serum creatinine levels and/or the duration of oliguria (Table 29.2).

As proposed, the first stratum provides the greatest sensitivity for diagnosing AKI, whereas higher strata provide increasing specificity of diagnosis. These three strata were combined with two outcome stages defined by the need for and duration of renal replacement therapy, resulting in the five-tiered RIFLE classification (**r**isk of renal dysfunction, **i**njury to the kidney, **f**ailure of kidney function, as well as the two outcome stages, **l**oss of kidney function and **e**nd-stage kidney disease).¹⁰ Subsequently, the Acute Kidney Injury Network (AKIN) modified the RIFLE classification by adding an absolute increase in serum creatinine level of 0.3 mg/dL or more to the 50% relative increase in the serum creatinine level to the definition of AKI and specifying that these increments develop over no more than 48 hours² (see Table 29.2). This definition was further modified in the Kidney Disease: Improving Global Outcomes (KDIGO) clinical practice guideline for acute kidney injury, which clarifies that although the 0.3-mg/dL increment in the serum creatinine level needs to occur over 48 hours, from a known baseline value, the 50% increase may occur over a longer, 7-day interval.³

The KDIGO clinical practice guideline for AKI recognized a gap in the nosology of acute and chronic kidney disease.³ Based on the definitions previously, AKI has an onset of less

Table 29.2 RIFLE, Acute Kidney Injury Network (AKIN) and Kidney Disease: Improving Global Outcomes (KDIGO) Definitions and Staging of Acute Kidney Injury

Definitions				
Parameter		RIFLE	AKIN	KDIGO
Serum creatinine		An increase of >50% developing over <7 days	An increase of >0.3 mg/dL or >50% developing over <48 hours	An increase of >0.3 mg/dL developing over <48 hours or an increase of >50% developing over <7 days
Urine output ^a		<0.5 mL/kg/h for >6 hours	<0.5 mL/kg/h for >6 hours	<0.5 mL/kg/h for >6 hours
Staging Criteria				
Increase in Serum Creatinine				
RIFLE		AKIN	KDIGO	Urine Output ^a
Risk	≥50%	Stage 1: ≥0.3 mg/dL or ≥50%	Stage 1: ≥0.3 mg/dL or ≥50%	<0.5 mL/kg/h for >6 hours
Injury	≥100%	Stage 2: ≥100%	Stage 2: ≥100%	<0.5 mL/kg/h for >12 hours
Failure	≥200%	Stage 3: ≥200%	Stage 3: ≥200%	<0.3 mL/kg/h for >24 hours or anuria for >12 hours
Loss	Need for renal replacement therapy for >4 weeks			
End-stage	Need for renal replacement therapy for >3 months			

^aThe urine output criteria for both definition and staging of AKI are the same for the RIFLE, AKIN, and KDIGO criteria.

RIFLE, **R**isk of renal dysfunction, **i**njury to the kidney, **f**ailure of kidney function, as well as the two outcome stages, **l**oss of kidney function and **e**nd-stage kidney disease.

than 7 days, whereas chronic kidney disease (CKD) is defined by the presence of impaired kidney function or structural kidney damage for more than 3 months with implications for health.¹¹ Recognizing that some patients develop kidney disease with a more subacute onset than that of AKI, but of less than 3 months' duration, the KDIGO Acute Kidney Injury Workgroup proposed the concept of acute kidney disease (AKD), defined as AKI or a reduction in the GFR to less than 60 mL/min/1.73 m², a decrease in the GFR by 35% or more, an increase in the serum creatinine level by more than 50%, or the presence of structural kidney damage of less than 3 months' duration.

Several limitations to these criteria for diagnosis and staging of AKI have been recognized.^{12,13} First, although validation studies have demonstrated that the AKI stage correlates with mortality risk, it is not clear that this is the appropriate metric for assessing their validity as a definition of kidney disease. Second, there is poor correlation between AKI stage and the GFR. Because the magnitude of change in the serum creatinine level is time-dependent, a patient may demonstrate progression over time from less severe (RIFLE-R, AKIN, or KDIGO stage 1) to more severe AKI stage (RIFLE-F, AKIN, or KDIGO stage 3), despite an improving GFR. Third, the definition of AKI by serum creatinine criteria relies on a referent baseline serum creatinine level, which is often unavailable. Furthermore, variations in specifications for this referent value (e.g., admission serum creatinine level vs. most recent outpatient serum creatinine level prior to admission vs. other definitions) can alter the classification of patients.¹⁴ Fourth, both RIFLE and AKIN definitions use relative changes in serum creatinine level to stage AKI. Analyses of creatinine kinetics have demonstrated that the time required to attain a fixed percentage change in serum creatinine level in the setting of severe AKI is dependent on the baseline level of kidney function, whereas the initial rate of change in serum creatinine level is relatively independent of kidney function.¹² Thus, early in the course of AKI, absolute changes in the serum creatinine level may be detected more readily than relative changes. Fifth, concordance between the serum creatinine level and urine output criterion is poor, even with regard to mortality risk.¹⁵ Transient changes in urine output may reflect variation in volume status or be due to the administration of medications and do not necessarily correlate with other parameters of kidney function. Finally, it must be noted that these classification systems are independent of the various causes of AKI—prerenal, intrinsic, obstructive. Despite these shortcomings, the use of standardized classification schemes has enhanced the interpretation of epidemiologic studies and the design of clinical trials.

Conceptually, AKI comprises a spectrum of structural and functional kidney disease in which there may be an evolution from injury to organ dysfunction and, finally, overt organ failure. Reliance solely on changes in serum creatinine level and/or urine output to diagnose AKI has resulted in the inability to identify the incipient stages of intrinsic kidney damage, which may be the most opportune time for pharmacologic intervention.¹⁶ To facilitate the early diagnosis of intrinsic injury, multiple biomarkers of tubular injury have been evaluated, including N-acetyl- β -D-glucosaminidase (NAG), kidney injury molecule 1 (KIM-1), neutrophil gelatinase-associated lipocalin (NGAL), interleukin 18 (IL-18), liver fatty acid binding protein (L-FABP), tissue inhibitor

of metalloproteinase 2 (TIMP-2), and insulin-like growth factor-binding protein 7 (IGFBP7).^{17–22} In addition, serum cystatin C has been proposed as more sensitive (and, in some settings, more specific) than serum creatinine for detecting changes in the GFR, and urinary cystatin C has been proposed as a marker of tubular injury.^{17,23,24} Although most of these biomarkers have yet to be adequately validated for routine clinical use, they have the potential to provide an earlier diagnosis of intrinsic AKI than the serum creatinine level, differentiate volume-responsive (prerenal) AKI from intrinsic disease, diminish the confounding effect related to creatinine generation, and provide prognostic information regarding the clinical course of an episode of AKI. One or more of these biomarkers may provide a means whereby patients could be identified at the incipient stage of AKI to guide the implementation of specific therapy to ameliorate kidney damage or promote recovery of kidney function.

INCIDENCE OF ACUTE KIDNEY INJURY

Estimates of the incidence of AKI are highly dependent on the case definition used, with rates among hospitalized patients ranging from as high as 44% when defined based on a change in serum creatinine level of at least 0.3 mg/dL to as low as 1% using an increase in serum creatinine level of at least 2.0 mg/dL.^{25–29} Approximately 3% to 7% of hospitalized patients and 25% to 60% of intensive care unit (ICU) patients develop AKI, with 5% to 6% of the ICU population requiring renal replacement therapy after developing AKI.^{25–29} In a single-center analysis conducted in 1996 at an urban tertiary care hospital, AKI, defined as an increase in serum creatinine level of 0.5 mg/dL for patients with a baseline serum creatinine level of 1.9 mg/dL or less, of 1.0 mg/dL for patients with a baseline serum creatinine level of 2.0 to 4.9 mg/dL, and 1.5 mg/dL for patients with a baseline serum creatinine level greater than 5 mg/dL, developed in 7.2% of 4622 consecutive patients.²⁵ The overall incidence of AKI is approximately 21.6% for all hospitalized adults worldwide,³⁰ with known associations for accelerating CKD to end-stage kidney disease (ESKD).^{31–33} The more recent estimate of AKI incidence is considerably higher than the 4.9% investigators had observed in a similar study in 1979.³⁴ In the late 1970s, the most frequent cause for AKI was decreased renal perfusion, observed in 39% of episodes, followed by medication-associated (16%), contrast-associated (11%), postoperative (9%), and sepsis-associated (6.5%). Overall mortality was 19.4%, with higher mortality rates associated with larger maximal increments in the serum creatinine concentration.

Although definition is less of an issue with regard to rates of AKI requiring renal replacement therapy, reported rates vary considerably because of differences in the characteristics of patient populations and variability in criteria for the initiation of renal replacement therapy. In a multinational, multicenter observational study of 29,269 critically ill patients, 5.7% developed severe AKI and 4.3% received renal replacement therapy (RRT).³⁵ In a smaller but more recent multinational observational study, 13.5% of 1802 critically ill patients developed RRT requiring AKI.²⁹

Many epidemiologic studies of AKI have relied on data from large administrative databases. Such data need to be

interpreted with caution, however, because administrative coding for AKI is incomplete and may only capture 20% to 30% of all episodes of AKI.^{36,37} The ascertainment of AKI requiring RRT using administrative data is substantially more complete. In an analysis of data from the National Hospital Discharge Survey in the United States, the Centers for Disease Control and Prevention have observed an increase in hospital discharges with a diagnosis of AKI from 18/100,000 population in 1980 to 365/100,000 in 2005.³⁸ Similar trends have been observed in analyses of the US Nationwide Inpatient Sample (NIS) and a 5% sample of US hospitalized Medicare beneficiaries. In an analysis that combined administrative and clinical data from a single integrated health care delivery system, the incidence of AKI that did not require the use of RRT increased from 322.7 to 522.4/100,000 person-years from 1996 to 2003.³⁹ Over the same period, AKI requiring RRT increased from 19.5 to 29.5/100,000 person-years.³⁹ AKI was more common among men and older adults. In a more recent analysis using data from the NIS, a nationally representative administrative database of hospitalizations, the incidence of dialysis-requiring AKI increased from 222/million person-years in 2000 to 533/million person-years in 2009, with the largest rise in incidence occurring in patients 65 to 74 years of age and 75 years of age or older.⁴⁰ Although a component of these temporal trends may be attributable to earlier initiation of dialysis and more frequent use of RRT in older patients, these changes are unlikely to account for most of the increase in incidence of severe AKI.

Preexisting kidney disease is one of the major risk factors for the development of dialysis-requiring AKI.⁴¹ More severe baseline CKD is associated with higher levels of risk. Compared with patients with a baseline estimated GFR (eGFR) more than 60 mL/min/1.73 m², patients with eGFR values of 45 to 59 mL/min/1.73 m² have a nearly twofold increased risk of developing dialysis-requiring AKI. This risk increases to more than 40-fold among patients with baseline eGFR values less than 15 mL/min/1.73 m².⁴¹ Underlying diabetes mellitus, hypertension, and the presence of proteinuria are also associated with the risk for hospital-acquired AKI.

CATEGORIZATION OF ACUTE KIDNEY INJURY

Although in the clinical setting AKI is often multifactorial, the cause is commonly evaluated based on three major pathophysiologic categories—prerenal, intrinsic, and postrenal (obstructive) AKI. Prerenal azotemia is the most common cause of AKI, accounting for about 40% to 55% of all cases.^{25,27,42} Prerenal AKI is caused by a reduction in blood flow to the kidneys that can result from absolute and/or effective reduction in intravascular volume. Common causes of absolute intravascular volume depletion leading to prerenal AKI include gastrointestinal losses (e.g., diarrhea, vomiting, nasogastric suction), renal losses (e.g., overdiuresis, diabetes insipidus), and sequestration of extracellular fluid (e.g., third spacing, as seen in acute pancreatitis; see [Box 29.1](#)). Common conditions that cause effective intravascular volume contraction include heart failure and liver failure. In these conditions, the absolute blood volume is increased, yet arterial blood flow to the kidneys is reduced, leading to prerenal azotemia.

A more detailed narrative on prerenal AKI is presented in Chapter 28 on the pathophysiology of AKI.

Intrinsic AKI is frequently categorized based on the anatomic location of primary renal injury, including the vasculature, glomeruli, interstitium, and tubules (see [Table 29.1](#)). Diseases that affect the renal vasculature range from complete thrombosis of the renal artery or vein, which are uncommon events, atheroembolic disease typically following instrumentation of renal or extrarenal vasculature, to vasculitides that affect small renal vessels, including the glomeruli. Such vasculitides may be confined to the kidney but are commonly systemic conditions with extrarenal manifestations. Acute interstitial nephritis (AIN) most often results from an idiosyncratic allergic response to a myriad of different pharmacologic agents, most commonly to antibiotics (e.g., methicillin and other penicillins, cephalosporins, sulfonamides, quinolones) or nonsteroidal antiinflammatory drugs (NSAIDs; e.g., ibuprofen).⁴³ Proton pump inhibitors have more recently been associated with impaired kidney function, hypothesized to be related, at least in part, to AIN.^{44–46} A multitude of other clinical conditions, including bacterial infection, can also lead to AIN. Systemic allergic signs such as fever, rash, and eosinophilia are often present in antibiotic-associated AIN, but not usually present in NSAID-related AIN.

The most common form of intrinsic AKI is acute tubular necrosis (ATN). The causes of ATN are broadly categorized as ischemic, septic, and nephrotoxic. Prolonged prerenal azotemia and hypotension, even if short-lived, can lead to necrosis of tubular epithelial cells and ATN. Nephrotoxins leading to ATN can be extrinsic, such as iodinated contrast, antibiotics, and chemotherapeutic agents, or intrinsic, such as myoglobin, hemoglobin, and intratubular crystals ([Table 29.3](#)). Sepsis-associated AKI, although previously attributed to ischemia reperfusion injury, is now believed to have a more complex pathogenesis and may develop in the absence of overt hypotension. These conditions, including their pathogenesis and pathophysiology, are discussed in detail in Chapter 28.

Postrenal (obstructive) AKI is characterized by blockage to the flow of urine. Obstruction, intrinsic or extrinsic, can occur at any point from the renal collecting system to the urethra (see [Box 29.2](#)). The development of postrenal AKI requires bilateral obstruction or unilateral obstruction in a patient with a solitary kidney or in the setting of CKD. The most common causes of postrenal AKI are functional or structural obstruction of the bladder neck due to prostatic conditions, anticholinergic agents, and neurogenic bladder. Additional causes of obstructive AKI include bilateral ureteral calculi, fibrosis, or blood clots, sloughed renal papillae (i.e., renal papillary necrosis), genitourinary malignancies, external compression from tumors or hemorrhage, and retroperitoneal fibrosis. A detailed description of renal obstruction is provided in Chapter 37.

EVALUATION OF ACUTE KIDNEY INJURY

The assessment of the patient with AKI requires a meticulous history and physical examination, comprehensive review of medical records, evaluation of urinary findings, including the urinary sediment, review of laboratory tests, renal imaging and, when appropriate, kidney biopsy⁴⁷ ([Table 29.4](#)). Analysis

Table 29.3 Major Causes of Endogenous and Exogenous Toxins Causing Acute Tubular Necrosis

Endogenous Toxins	Exogenous Toxins
Myoglobinuria - Muscle breakdown—trauma, compression, electric shock, hypothermia, hyperthermia, seizures, exercise, burns - Metabolic—hypokalemia, hypophosphatemia - Infections—tetanus, influenza, - Toxins—isopropyl alcohol, ethanol, ethylene glycol, toluene, snake and insect bites, cocaine, heroin - Drugs—HMG-CoA reductase inhibitors (statins), amphetamines, fibrates - Inherited disease—deficiency of myophosphorylase, phosphofructokinase, carnitine palmitoyltransferase - Autoimmune—polymyositis, dermatomyositis Hemoglobinuria - Mechanical—prosthetic valves, microangiopathic hemolytic anemia, extracorporeal circulation - Drugs—hydralazine, methyl dopa - Chemicals—benzene, arsine, fava beans, glycerol, phenol - Immunologic—transfusion reaction - Genetic—G6PD deficiency, PNH Intratubular obstruction from crystalluria or paraproteins - Tumor lysis syndrome - HGPT deficiency Multiple myeloma - Oxalate (ethylene glycol)	Antibiotics - Aminoglycosides - Amphotericin B - Antiviral agents—acyclovir, cidofovir, indinavir, foscarnet, tenofovir - Pentamidine - Vancomycin Chemotherapy - Cisplatin - Ifosfamide - Plicamycin 5-Fluorouracil - Cytarabine 6-Thioguanine - Methotrexate Calcineurin inhibitors - Cyclosporine - Tacrolimus Organic solvents - Toluene - Ethylene glycol Poisons Miscellaneous - Radiocontrast agents - Intravenous immune globulin - Nonsteroidal antiinflammatory drugs - Oral phosphate bowel preparations

G6PD, Glucose-6-phosphate dehydrogenase; *HGPT*, Hypoxanthine-guanine phosphoribosyltransferase; *HMG-CoA*, 5-hydroxy-3-methylglutaryl-coenzyme A; *PNH*, paroxysmal nocturnal hemoglobinuria.

of serum creatinine concentration over time is invaluable for differentiating acute from chronic kidney disease and identifying the timing of events that precipitated the acute decline in kidney function. The presence of an acute process is easily confirmed if review of laboratory records reveals a sudden rise in blood urea nitrogen (BUN) level and serum creatinine concentration from previously stable baseline values. Spurious causes of elevated BUN and serum creatinine levels must be excluded before a diagnosis of AKI is made. When prior BUN and serum creatinine measurements are not available, key findings that suggest that a chronic process is present include physical manifestations of hyperparathyroidism (e.g., resorption of distal phalangeal tufts, lateral aspect of clavicles), band keratopathy, so-called half-and-half nails, and small echogenic kidneys on radiographic imaging. Enlarged kidneys do not necessarily rule out a chronic process because diabetic nephropathy, HIV-associated nephropathy, amyloidosis, and polycystic kidney disease are characterized by enlarged kidney size, even with moderate to advanced CKD. Anemia is a less useful differentiating feature because it is often present in both AKI and CKD. Once the presence of AKI has been confirmed, attention should focus on patient, urine, laboratory, and radiographic assessments to

help distinguish among prerenal, intrinsic, and postrenal processes, permit identification of the cause of AKI, and guide treatment.

CLINICAL ASSESSMENT OF THE PATIENT

Prerenal AKI should be suspected in clinical settings associated with intravascular volume depletion, including hemorrhage, excessive gastrointestinal, urinary, or insensible fluid losses and severe burns, or with reduced effective arterial blood volume due to congestive heart failure, liver disease, or nephrotic syndrome. The risk of intravascular volume depletion is increased in comatose, sedated, or obtunded patients and in patients with restricted access to salt and water. Clinical clues to a prerenal cause of AKI on history include patient report of excessive thirst, orthostatic light-headedness or dizziness, significant diarrhea and/or vomiting, diuretic use, and recent use of medications that alter intrarenal hemodynamics, including NSAIDs, as well as inhibitors of the renin-angiotensin axis, such as direct renin inhibitors, angiotensin-converting enzyme (ACE) inhibitors, and angiotensin receptor blockers (ARBs).

Findings suggestive of volume depletion on physical examination may include orthostatic hypotension (postural

Table 29.4 Useful Clinical Features, Urinary Findings, and Confirmatory Tests in the Differential Diagnosis of Acute Kidney Injury

Cause of Acute Kidney Injury	Some Suggestive Clinical Features	Typical Urinalysis	Some Confirmatory Tests
Prerenal azotemia	Evidence of true volume depletion (thirst, postural or absolute hypotension and tachycardia, low jugular venous pressure, dry mucous membranes and axillae, weight loss, fluid output >input) or decreased effective circulatory volume (e.g., heart failure, liver failure), treatment with NSAID, diuretic, or ACE inhibitor or ARB	Hyaline casts $FE_{Na} < 1\%$ $U_{Na} < 10$ mmol/L $SG > 1.018$	Occasionally requires invasive hemodynamic monitoring; rapid resolution of AKI with restoration of renal perfusion
Diseases Involving Large Renal Vessels			
Renal artery thrombosis	History of atrial fibrillation or recent myocardial infarction, nausea, vomiting, flank or abdominal pain	Mild proteinuria Occasionally RBCs	Elevated LDH with normal transaminase levels, renal arteriogram, MAG-3 renal scan, MRA ^a
Atheroembolism	Usually age >50 years, recent manipulation of aorta, retinal plaques, subcutaneous nodules, palpable purpura, livedo reticularis	Often normal Eosinophiluria Rarely casts	Eosinophilia, hypocomplementemia, skin biopsy, renal biopsy
Renal vein thrombosis	Evidence of nephrotic syndrome or pulmonary embolism, flank pain	Proteinuria, hematuria	Inferior venocavogram, Doppler flow studies, MRV ^a
Diseases of Small Renal Vessels and Glomeruli			
Glomerulonephritis or vasculitis	Compatible clinical history (e.g., recent infection), sinusitis, lung hemorrhage, rash or skin ulcers, arthralgias, hypertension, edema	RBC or granular casts, RBCs, white blood cells, proteinuria	Low complement levels; positive antineutrophil cytoplasmic antibodies, antiglomerular basement membrane antibodies, antistreptolysin O antibodies, anti-DNAse, cryoglobulins; renal biopsy
HUS, TTP	Compatible clinical history (e.g., recent gastrointestinal infection, cyclosporine, anovulants), pallor, ecchymoses, neurologic findings	May be normal, RBCs, mild proteinuria, rarely RBC or granular casts	Anemia, thrombocytopenia, schistocytes on peripheral blood smear, low haptoglobin, increased LDH, renal biopsy
Malignant hypertension	Severe hypertension with headaches, cardiac failure, retinopathy, neurologic dysfunction, papilledema	May be normal, RBCs, mild proteinuria, rarely RBC casts	LVH by echocardiography or EKG, resolution of AKI with BP control
Ischemic or Nephrotoxic Acute Tubular Necrosis			
Ischemia	Recent hemorrhage, hypotension, surgery, often in combination with vasoactive medication (e.g., ACE inhibitor, NSAID)	Muddy brown granular or tubular epithelial cell casts $FE_{Na} > 1\%$, $U_{Na} > 20$ mmol/L, $SG \approx 1.010$	Clinical assessment and urinalysis usually inform diagnosis.
Exogenous toxin	Recent contrast-enhanced procedure; nephrotoxic medications; certain chemotherapeutic agents often with coexistent volume depletion, sepsis or chronic kidney disease	Muddy brown granular or tubular epithelial cell cases $FE_{Na} > 1\%$, $U_{Na} > 20$ mmol/L, $SG \approx 1.010$	Clinical assessment and urinalysis usually inform diagnosis.
Endogenous toxin	History suggestive of: Rhabdomyolysis (coma, seizures, drug abuse, trauma) Hemolysis (recent blood transfusion) Tumor lysis (recent chemotherapy) Myeloma (bone pain) or ethylene glycol ingestion	Urine supernatant tests positive for heme in absence of red cells; urine supernatant pink and tests positive for heme in absence of red cells; urate crystals, dipstick-negative proteinuria, oxalate crystals, respectively	Hyperkalemia, hyperphosphatemia, hypocalcemia, increased CK, myoglobin; hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia and free circulating hemoglobin; hyperuricemia, hyperkalemia, hyperphosphatemia (for tumor lysis); circulating or urinary monoclonal protein (for myeloma); toxicology screen, acidosis, osmolal gap (for ethylene glycol)

Table 29.4 Useful Clinical Features, Urinary Findings, and Confirmatory Tests in the Differential Diagnosis of Acute Kidney Injury (Cont'd)

Cause of Acute Kidney Injury	Some Suggestive Clinical Features	Typical Urinalysis	Some Confirmatory Tests
Diseases of the tubulointerstitium— allergic interstitial nephritis	Recent ingestion of drug and fever, rash, loin pain, or arthralgias	White blood cell casts, white blood cells (frequently eosinophiluria), RBCs, rarely RBC casts, proteinuria (occasionally nephritic)	Systemic eosinophilia, renal biopsy
Acute bilateral pyelonephritis	Fever, flank pain and tenderness, toxic state	Leukocytes, occasionally white blood cell casts, RBCs, bacteria	Urine and blood cultures
Postrenal AKI	Abdominal and flank pain, palpable bladder	Frequently normal, hematuria if stones, prostatic hypertrophy	Plain abdominal x-ray, renal ultrasonography, postvoid residual bladder volume, computed tomography, retrograde or antegrade pyelography

^aContrast-enhanced magnetic resonance angiography and magnetic resonance venography should be used with extreme caution in patients with AKI.

ACE, Angiotensin-converting enzyme; AKI, acute kidney injury; ARB, angiotensin receptor blocker; BP, blood pressure; CK, creatine kinase; EKG, electrocardiogram; FE_{Na} , fractional excretion of sodium; HUS, hemolytic-uremic syndrome; LDH, lactate dehydrogenase; LVH, left ventricular hypertrophy; MRA, magnetic resonance angiography; MRV, magnetic resonance venography; NSAID, nonsteroidal antiinflammatory drug; RBC, red blood cell; SG, specific gravity; TTP, thrombotic thrombocytopenic purpura; U_{Na} , urine sodium.

fall in diastolic blood pressure >10 mm Hg) and tachycardia (postural increase in heart rate >10 beats/min), reduced jugular venous pressure, diminished skin turgor, dry mucous membranes, and the absence of axillary sweat. However, overt signs and symptoms of hypovolemia do not usually manifest until the extracellular fluid volume has fallen by more than 10% to 20%. In addition, in patients with heart failure, liver disease, or nephrotic syndrome, renal hypoperfusion may be present, despite total body volume overload. Findings on physical examination of peripheral edema, pulmonary vascular congestion, pleural effusion, cardiomegaly, the presence of an S3 heart sound, elevated jugular venous pressure, or hepatic congestion may point to a state of reduced cardiac output and decreased effective intravascular volume. The presence of acute or chronic liver disease is suggested by evidence of icterus, ascites, splenomegaly, palmar erythema, telangiectasia, and caput medusa. In select critically ill patients, invasive hemodynamic monitoring using central venous or pulmonary artery catheters or ultrasonography of the heart and central veins may assist in assessing intravascular volume status. A definitive diagnosis of prerenal AKI is usually based on the prompt resolution of AKI after restoration of renal perfusion. In patients with underlying systolic heart failure, restoration of renal perfusion may be challenging and often requires the use of inotropic support.

There is a high likelihood of ischemic ATN if AKI follows a period of severe renal hypoperfusion, and the impairment in kidney function persists or worsens, despite restoration of renal perfusion. It should be noted, however, that significant hypotension is evident in fewer than 50% of patients with postsurgical ATN.⁴⁸ Although septic shock is a common cause of ATN, ATN may also develop in sepsis in the absence of overt hypotension.^{48,49} The diagnosis of nephrotoxic ATN

requires a comprehensive review of all clinical, pharmacy, nursing, radiographic, and procedural notes for evidence of the administration of nephrotoxic agents. Pigment-induced ATN may be suspected if the clinical assessment reveals risk factors for rhabdomyolysis (e.g., seizures, excessive exercise, alcohol or drug abuse, treatment with statins, prolonged immobilization, limb ischemia, crush injury) or hemolysis, as well as select signs and symptoms of the former (e.g., muscle tenderness, weakness, evidence of trauma or prolonged immobilization).^{50–53}

Although most AKI is prerenal or ischemic, nephrotoxic, or septic ATN, patients should be carefully evaluated for other intrinsic renal parenchymal processes because their management and prognosis may differ substantially. Flank pain may be a prominent symptom of acute renal artery or renal vein occlusion, renal infarction (from systemic emboli), acute pyelonephritis and, rarely, necrotizing glomerulonephritis.^{54–56} Interstitial edema leading to distention of the renal capsule and flank pain may be seen in up to one-third of patients with AIN.⁵⁷ Dermatologic examination can be highly informative because a maculopapular rash may accompany allergic interstitial nephritis: subcutaneous nodules, livedo reticularis, digital ischemia, and palpable purpura may suggest atheroembolism or vasculitis; a malar (butterfly) rash may be associated with systemic lupus erythematosus; and impetigo or needle tracks from intravenous (IV) drug use may underlie infection-associated glomerulonephritis. An ophthalmologic examination is useful to assess for signs of atheroembolism; hypertensive or diabetic retinopathy; the keratitis, uveitis, and iritis of autoimmune vasculitides; icterus associated with chronic liver disease; and the rare but nevertheless pathognomonic band keratopathy of hypercalcemia and flecked retina of hyperoxalemia. Uveitis may also

be an indicator of coexistent allergic interstitial nephritis, sarcoidosis, and the tubulointerstitial nephritis and uveitis (TINU) syndrome.⁵⁸ Examination of the ears, nose, and throat may reveal conductive deafness and mucosal inflammation or ulceration suggestive of necrotizing granulomatous vasculitis or the neural deafness caused by aminoglycoside toxicity. Respiratory failure, particularly if associated with hemoptysis, suggests the presence of a pulmonary-renal syndrome, and ascites and other stigmata of severe chronic liver disease suggest the possibility of hepatorenal syndrome (HRS). Cardiovascular assessment may reveal marked elevation in systemic blood pressure suggesting malignant hypertension or scleroderma or may demonstrate a new arrhythmia or murmur suggesting a potential source of thromboemboli or subacute bacterial endocarditis (acute glomerulonephritis), respectively. Chest or abdominal pain and reduced pulses in the lower limbs should suggest aortic dissection or, rarely, Takayasu arteritis. Abdominal pain and nausea are frequent clinical correlates of atheroembolic disease, commonly in patients who have recently undergone angiographic evaluation, particularly in the presence of widespread atherosclerotic disease. The presence of a tensely distended abdomen may indicate the presence of abdominal compartment syndrome and should prompt transduction of bladder pressure.⁵⁹ Pallor and recent bruising are important clues to the thrombotic microangiopathies, and the combination of bleeding and fever should raise the possibility of AKI resulting from viral hemorrhagic fevers. A recent jejunoileal bypass may be a vital clue to oxalosis, a rare but reversible cause of AKI following bariatric surgery.⁶⁰ Hyperreflexia and asterixis often portend the development of uremic encephalopathy or may, in the presence of focal neurologic signs, suggest a diagnosis of thrombotic microangiopathy (i.e., hemolytic-uremic syndrome [HUS] or thrombotic thrombocytopenic purpura [TTP]; see Chapter 34).

Postrenal AKI may be asymptomatic if obstruction to the drainage of urine develops gradually. Although anuria will be seen in complete obstruction, urine volume may be normal or even increased in the setting of partial obstruction. A pattern of fluctuating urine output may also be seen in some patients with partial obstruction. Suprapubic or flank pain may be the presenting complaint if there is acute distention of the bladder or renal collecting system and capsule, respectively. Colicky flank pain radiating to the groin suggests acute ureteral obstruction, most commonly from urinary stone disease. Prostatic disease should be suspected in older men with a history of nocturia, urinary frequency, urgency or hesitancy, and an enlarged prostate on rectal examination. Urinary retention may be exacerbated acutely in such patients by medications with anticholinergic properties, such as antihistamines or antidepressants. Rectal or pelvic examination may reveal obstructing tumors in female patients. Neurogenic bladder is a likely diagnosis in patients with spinal cord injury or autonomic insufficiency and should be suspected in patients with long-standing diabetes mellitus. Bladder distention may be evident on abdominal percussion and palpation in patients with bladder neck or urethral obstruction. Definitive diagnosis of postrenal AKI usually relies on examination of the postvoid bladder volume and radiographic evaluation of the upper urinary tract and is confirmed by improvement in kidney function following relief of the obstruction.

URINE ASSESSMENT

Urine volume is rarely helpful in distinguishing various forms and causes of AKI. Anuria can be seen with complete urinary tract obstruction but can also be seen with severe prerenal or intrinsic renal disease (e.g., renal artery occlusion, severe proliferative glomerulonephritis or vasculitis, bilateral cortical necrosis). Patients with partial urinary tract obstruction may present with polyuria caused by secondary impairment of urine-concentrating mechanisms.

Assessment of the urine is essential in patients with AKI and is an inexpensive and useful diagnostic tool.^{61–63} Measured urine specific gravity above 1.015 to 1.020 often accompanies prerenal AKI, although impaired urinary concentration may be present in patients with underlying CKD or as a result of diuretic therapy. Acute glomerulonephritis may also present with concentrated urine. Isosthenuria (a urine specific gravity of 1.010, similar to that of plasma) is characteristic of ATN. Hematuria on dipstick assessment may result from urologic trauma from catheterization, urologic disease, interstitial nephritis, acute glomerulonephritis, atheroembolic disease, renal infarction, or pigment (hemoglobinuric or myoglobinuric) nephropathy. The latter are suggested when the dipstick test for blood is strongly positive but there are few or no red blood cells seen on microscopic examination of the urinary sediment.

Examination of the urinary sediment of a centrifuged urine specimen complements the dipstick analysis and is highly valuable for distinguishing among the various forms of AKI (Box 29.3). The sediment should be inspected for the presence of cells, casts, and crystals. In prerenal AKI, the urine sediment is typically bland (i.e., devoid of cells or casts) but may contain transparent hyaline casts. Hyaline casts are formed in concentrated urine from normal urinary constituents, principally Tamm-Horsfall protein secreted by epithelial cells of the loop of Henle. Postrenal AKI may also present with a bland urine sediment, although hematuria is common in patients with intraluminal obstruction (e.g., stones, sloughed papilla, blood clot) or prostatic disease. Renal tubular epithelial cells, epithelial cell casts, and pigmented, muddy brown granular casts are characteristic of ischemic or nephrotoxic ATN. These characteristic findings of ATN may be found in association with microscopic hematuria and mild tubular proteinuria (<1 g/day). Casts may be absent in approximately 20% to 30% of patients with ischemic or nephrotoxic ATN and are not a requisite for diagnosis^{61,64}; however, semiquantitative scoring systems have been developed to assess the presence of epithelial cells and casts in patients with AKI to assist in the diagnosis of ATN and correlate with the clinical course.^{62,65,66}

Red blood cell (RBC) casts are almost always indicative of acute glomerular disease but may be observed, albeit rarely, in AIN. Dysmorphic RBCs, best seen using phase contrast microscopy, are a more common urinary finding in patients with glomerular injury but are a less specific finding than RBC casts. Urine sediment abnormalities vary in diseases involving preglomerular blood vessels, such as HUS, TTP, atheroembolic disease, and vasculitis involving medium-sized or large vessels, and range from benign to overtly nephritic. White blood cell casts and nonpigmented granular casts suggest interstitial nephritis, whereas broad granular casts are characteristic of CKD and probably reflect interstitial fibrosis and dilation of

Box 29.3 Urine Sediment in the Differential Diagnosis of Acute Kidney Injury

Normal or few red blood cells or white blood cells
 Prerenal azotemia
 Arterial thrombosis or embolism
 Proliferative glomerulonephritis
 HUS, TTP
 Scleroderma crisis
 Postrenal AKI
 Renal tubular epithelial cells and granular casts
 Acute tubular necrosis
 Dysmorphic red blood cells and red blood cell casts
 Glomerulonephritis or vasculitis
 Malignant hypertension
 Rarely interstitial nephritis
 White blood cell and white blood cell casts
 Acute interstitial nephritis or exudative glomerulonephritis
 Severe pyelonephritis
 Marked leukemic or lymphomatous infiltration
 Eosinophiluria (>5%)
 Allergic interstitial nephritis (antibiotics >> NSAIDs)
 Atheroembolism
 Crystalluria
 Acute urate nephropathy
 Calcium oxalate (ethylene glycol intoxication)
 Acyclovir
 Indinavir
 Sulfonamides
 Methotrexate

AKI, Acute kidney injury; HUS, hemolytic-uremic syndrome; NSAIDs, nonsteroidal antiinflammatory drugs; TTP, thrombotic thrombocytopenic purpura.

tubules. Eosinophiluria (from 1%–50% of urine leukocytes) is a common finding (90%) in drug-induced allergic interstitial nephritis.^{67,68} However, eosinophiluria has poor sensitivity and specificity for the diagnosis of AIN, with eosinophiluria of 1% to more than 5% occurring in a variety of other diseases, including atheroembolization, ischemic and nephrotoxic AKI, proliferative glomerulonephritis, pyelonephritis, cystitis, and prostatitis. In a series of 566 patients who had urinary eosinophil testing and renal histology from kidney biopsy, eosinophiluria only had a 31% sensitivity and 68% specificity for the diagnosis of interstitial nephritis.⁶⁹ Uric acid crystals (pleomorphic) may be seen in the urine in prerenal AKI but should raise the possibility of acute urate nephropathy if seen in abundance. Oxalate crystalluria (needle- or dumbbell-shaped monohydrate crystals or envelope-shaped dihydrate crystals) may suggest a diagnosis of ethylene glycol toxicity.⁷⁰ A variety of other crystals may be seen in other medication-associated crystal nephropathies.^{71–73}

Increased urinary protein excretion, characteristically less than 1 g/day, is a common finding in ischemic or nephrotoxic ATN and reflects both failure of injured proximal tubule cells to reabsorb normally filtered protein and excretion of cellular debris (tubular proteinuria). Proteinuria greater than 1 g/day suggests injury to the glomerular ultrafiltration barrier (glomerular proteinuria) or excretion of light chains.^{74,75}

The latter are not detected by conventional dipsticks, which detect albumin, and must be sought by other means (e.g., sulfosalicylic acid test). Heavy proteinuria is also a frequent finding (80%) in patients with allergic interstitial nephritis triggered by NSAIDs. In addition to acute interstitial inflammation, these patients have a glomerular lesion that is almost identical to that of minimal-change disease.⁷⁶ A similar syndrome has been reported in patients receiving other agents, such as ampicillin, rifampin, and interferon alfa.^{77,78} Hemolysis and rhabdomyolysis may often be differentiated by inspection of plasma, which is characteristically pink in hemolysis, but clear in rhabdomyolysis.

Analysis of urine biochemical parameters may be helpful in differentiating between prerenal and intrinsic ischemic or nephrotoxic ATN (Table 29.5). Sodium is usually avidly reabsorbed from the glomerular filtrate in patients with prerenal AKI as a consequence of renal adrenergic activation, stimulation of the renin-angiotensin-aldosterone system (RAAS), suppression of atrial natriuretic peptide (ANP) secretion, and local changes in peritubular hemodynamics. In contrast, Na^+ reabsorption is impaired in ATN as a result of injury to the renal tubular epithelium. Renal sodium handling can be assessed based on the urinary sodium concentration (U_{Na}), with values of less than 10 mmol/L commonly seen in prerenal disease compared with more than 20 mmol/L in ATN. Normalizing sodium excretion to creatinine provides a more sensitive index. The fractional excretion of sodium (FE_{Na}) is the ratio between urine sodium excretion ($U_{\text{Na}} \times V$, where U_{Na} is the urine sodium concentration and V is the urine volume) and the filtered load of sodium (calculated as $P_{\text{Na}} \times \text{CrCl}$, where P_{Na} is the plasma sodium concentration and CrCl is the creatinine clearance, which can be calculated as $[(U_{\text{Cr}} \times V)/P_{\text{Cr}}]$ where V is the urine volume and $[U_{\text{Cr}}]$ and $[P_{\text{Cr}}]$ are the urine and plasma creatinine concentrations, respectively. Because urine volume is in both the numerator and denominator of this ratio, the FE_{Na} can be calculated as $[(U_{\text{Na}} \div P_{\text{Na}})/(U_{\text{Cr}} \div P_{\text{Cr}})] \times 100$ using an untimed (spot) urine sample and simultaneous serum sodium and creatinine measurements; this is usually less than 1% (frequently, <0.5%) in the setting of prerenal azotemia, whereas it is typically more than 2% in patients with ischemic or nephrotoxic AKI.

The utility of the FE_{Na} is limited in a variety of clinical settings. Values greater than 1% are not uncommon in the setting of prerenal AKI in patients receiving diuretics, those with metabolic alkalosis and bicarbonaturia (in whom Na^+ is excreted with HCO_3^- to maintain electroneutrality), in the presence of adrenal insufficiency, and in the setting of underlying CKD.^{79–81} Conversely, an FE_{Na} less than 1% may be observed in the setting of ATN, particularly in the settings of iodinated contrast administration, rhabdomyolysis, and sepsis, although it has been reported in 15% or more of patients with ATN from a variety of other causes, including ischemia, burns, and exposure to selected nephrotoxins.^{79,80,82–84}

It has been postulated that a low FE_{Na} value reflects a milder degree of intrinsic renal injury in which epithelial cell damage is probably localized to the corticomedullary junction and outer medulla, with relative preservation of function in other Na^+ -transporting segments, and may represent a transition state between prerenal azotemia and ATN. It should be recognized that an FE_{Na} value of less than 1% is not abnormal and reflects normal sodium homeostasis in

Table 29.5 Urine Indices Used in the Differential Diagnosis of Prerenal Acute Kidney Injury and Acute Tubular Necrosis

Diagnostic Index	Prerenal AKI	ATN
Fractional excretion of sodium (%)	<1 ^a	>2 ^a
Urine sodium concentration (mmol/L)	<20	>40
Urine creatinine-to-plasma creatinine ratio	>40	<20
Urine urea nitrogen-to-plasma urea nitrogen ratio	>8	<3
Urine specific gravity	>1.018	~1.010
Urine osmolality (mOsm/kg H ₂ O)	>500	~300
Plasma BUN-to-creatinine ratio	>20	<10-15
Renal failure index, $U_{Na}/(U_{Cr}/P_{Cr})$	<1	>1
Urine sediment	Hyaline casts	Muddy brown granular casts

^aFE_{Na} may be >1% in prerenal AKI associated with diuretic use and/or the setting of bicarbonaturia or chronic kidney disease; FE_{Na} often <1% in acute tubular necrosis caused by radiocontrast or rhabdomyolysis.

AKI, Acute kidney injury; ATN, acute tubular necrosis; BUN, blood urea nitrogen; P_{Cr}, plasma creatinine; U_{Cr}, urine creatinine; U_{Na}, urine sodium.

patients on a moderate- to low-sodium diet. The FE_{Na} is also often less than 1% in AKI caused by urinary tract obstruction, glomerulonephritis, and diseases of the renal vasculature; other parameters must be used to distinguish these conditions from prerenal AKI.

A variety of other indices has also been proposed to differentiate between causes of AKI. The renal failure index, calculated as $U_{Na}/(U_{Cr} + P_{Cr})$ provides comparable information to the FE_{Na} because clinical variations in serum Na⁺ concentration are relatively small. The fractional excretion of urea (FE_{urea}) has been proposed as an alternative to the FE_{Na}, with particular utility in patients on diuretic therapy. Values of FE_{urea} calculated as $([U_{urea} + P_{urea}]/[U_{Cr} + P_{Cr}] \times 100)$ less than 35% are suggestive of a prerenal state.⁸⁵⁻⁸⁷ Similarly, indices of urinary concentrating ability, such as urine specific gravity, urine osmolality, urine-to-plasma creatinine or urea ratios, and serum urea nitrogen-to-creatinine ratio are of limited value in differentiating between prerenal and intrinsic AKI. This is particularly true for older patients, in whom urine concentrating mechanisms are frequently impaired while mechanisms for Na⁺ reabsorption are typically preserved.

BLOOD AND LABORATORY FINDINGS

The pattern and timing of changes of BUN and serum creatinine levels often provide clues to the cause of AKI. Enhanced tubular reabsorption of filtered urea, in parallel with sodium and water reabsorption in prerenal states, commonly leads to a disproportionate elevation in BUN relative to serum creatinine levels (ratio >20:1). Conversely, with intrinsic AKI, the increase in BUN level usually parallels the rise in serum creatinine level, maintaining a ratio of about 10:1. However, severe malnutrition and low dietary protein intake blunt the rise in BUN and creatinine levels, whereas gastrointestinal bleeding, steroid therapy, and hypercatabolic states may lead to increases in the BUN level that do not reflect prerenal physiology. In addition, aggressive volume resuscitation may rapidly expand the volume of distribution of urea and creatinine and may also obscure the acute rise in serum creatinine level. Sepsis and other forms of critical

illness have also been associated with decreased creatinine generation.^{88,89} The serum creatinine level typically begins to rise within 24 to 48 hours when ATN results from an ischemic insult. Although the clinical course can be highly variable, the serum creatinine level will generally peak within 7 to 10 days and, depending on the severity of the insult and underlying comorbid illnesses, AKI will resolve to varying degrees over the ensuing 1 to 2 weeks. Following iodinated contrast exposure, the peak in serum creatinine level generally occurs within 5 to 7 days. The time course of nephrotoxic ATN caused by aminoglycoside antibiotics or cisplatin is more variable, often with a delayed onset of AKI (7–10 days).

Additional clues to the diagnosis can be obtained from biochemical and hematologic tests. The presence of marked hyperkalemia, hyperuricemia, and hyperphosphatemia point to cell lysis, which, in the setting of elevated creatine kinase levels and hypocalcemia, strongly suggests rhabdomyolysis.^{90,91} Biochemical signs of cell lysis with very high levels of uric acid, normal or mildly elevated creatine kinase levels, and a urine uric acid-to-creatinine ratio greater than 1.0 are suggestive of acute urate nephropathy and tumor lysis syndrome.^{92,93} Severe hypercalcemia can precipitate AKI, commonly in the form of prerenal AKI from concomitant hypovolemia and renal vasoconstriction. AKI associated with a widening of both the serum anion ($Na^+ - [HCO_3^- + Cl^-]$) and osmolal (measured serum osmolality minus calculated osmolality) gaps suggests a diagnosis of ethylene glycol toxicity and should prompt a search for urine oxalate crystals. Severe anemia in the absence of hemorrhage may reflect the presence of hemolysis, multiple myeloma, or thrombotic microangiopathy (e.g., HUS, TTP, toxemia, disseminated intravascular coagulation, accelerated hypertension, systemic lupus erythematosus [SLE], scleroderma, radiation injury). Other laboratory findings suggestive of thrombotic microangiopathy include thrombocytopenia, dysmorphic RBCs on peripheral blood smear, low circulating haptoglobin concentrations, and elevated circulating concentrations of lactate dehydrogenase. Systemic eosinophilia suggests allergic interstitial nephritis but may also be a prominent feature in other diseases, such as atheroembolic disease and eosinophilic

granulomatosis with polyangiitis (Churg-Strauss disease). Depressed serum complement and high titers of antglomerular basement membrane antibodies, antineutrophil cytoplasmic antibodies, antinuclear antibodies, circulating immune complexes, or cryoglobulins are useful diagnostic tools in patients with suspected glomerulonephritis or systemic vasculitis.

NOVEL BIOMARKERS OF KIDNEY INJURY

A number of novel biomarkers of kidney injury have been evaluated for potential roles in the early identification, differential diagnosis, and prognosis of AKI, including serum cystatin C, NGAL, KIM-1, IL-18, L-FABP, TIMP-2 and IGFBP7 among others (see earlier).^{17–24} Although most of these biomarkers are not available for routine clinical use, they have the potential to provide an earlier diagnosis of intrinsic AKI, differentiate volume-responsive (prerenal) AKI from intrinsic disease, and provide prognostic information regarding the clinical course of an episode of AKI.

Cystatin C

Cystatin C is a 13-kDa protein that is filtered by the glomerulus and completely reabsorbed and degraded by the proximal tubule. Cystatin C has been validated as an alternative marker of glomerular filtration.^{94,95} As a result of its shorter serum half-life, serum cystatin C concentrations change more rapidly than serum creatinine levels in response to changes in kidney function, allowing changes in serum cystatin C to be detected sooner than changes in serum creatinine following the onset of AKI.⁹⁶ Under normal circumstances, urinary cystatin C is virtually undetectable; however, following tubular injury, tubular reabsorption of filtered cystatin is diminished, and urinary cystatin C can be detected, raising the possibility of its use as an early marker of tubular injury.⁹⁷

Neutrophil Gelatinase-Associated Lipocalin

NGAL is a 25-kDa protein whose expression by renal tubular epithelial cells is markedly upregulated following ischemic or nephrotoxic kidney injury.^{98,99} NGAL is believed to enhance the trafficking of iron-siderophore complexes, enhancing the delivery of iron, upregulating hemeoxygenase-1, reducing apoptosis, and increasing the normal proliferation of renal tubule epithelial cells.¹⁰⁰ Urine and plasma NGAL have been evaluated in numerous clinical settings as an early biomarker of tubular injury.^{101–111} Initial studies in children undergoing cardiac surgery have demonstrated extremely high sensitivity and specificity for the identification of AKI, with an area under the receiver operating characteristic (ROC) curve of more than 0.99; however, these early results have not been reproduced across other clinical settings.

Kidney Injury Molecule 1

KIM-1 is a transmembrane protein whose expression is markedly upregulated in the proximal tubule following tubular injury.^{112–115} The extracellular component of the KIM-1 protein is shed into the urine following tubular injury, permitting its potential use as a marker of tubular damage; however, the time course of peak KIM-1 expression in the urine is later than that seen with NGAL.^{108,116–118} Moreover, more recent studies have demonstrated elevated levels of KIM-1 in non-AKI settings, including chronic kidney disease and renal cell carcinoma, which reduce its specificity for AKI.^{119,120}

Interleukin 18

IL18 is a proinflammatory cytokine whose expression is increased in the kidney following ischemic and nephrotoxic injury.¹²¹ Urinary IL-18 levels have been shown to rise within 6 hours following tubular injury, following cardiac surgery, and in critically ill patients.^{110,111,122,123}

Liver Fatty Acid-Binding Protein

Despite its name, L-FABP is expressed in the proximal tubule.^{124,125} Elevated L-FABP levels may be detected in the urine within 6 hours of ischemic or nephrotoxic injury, permitting its potential use as a marker of tubular injury.^{118,126–128} In a meta-analysis of published studies, the sensitivity and specificity of urinary L-FABP for diagnosis of AKI were each approximately 75%.¹²⁹

Tissue Inhibitor of Metalloproteinase 2 and Insulin-Like Growth Factor-Binding Protein 7

TIMP-2 and IGFBP7 are expressed in epithelial cells and act in an autocrine and paracrine manner to arrest cell cycle in AKI.^{130–132} In three discovery cohorts comprising 522 patients, these two biomarkers were identified as having the highest discriminant ability among 340 candidate biomarkers of AKI.²² In a subsequent validation study of 728 patients, this pair of biomarkers had an area under the ROC curve of 0.80, which was significantly better than the performance of other candidate biomarkers, including NGAL, KIM-1, IL-18, and L-FABP.²² A combination test that includes TIMP-2 and IGFBP7 is available for commercial use.

RADIOLOGIC EVALUATION

Imaging of the abdomen is a highly useful adjunct to laboratory testing to determine the cause of AKI. In cases of suspected obstructive uropathy, postvoid residual volumes of more than 100 to 150 mL suggest a diagnosis of bladder outlet obstruction. Although plain films rarely provide definitive evidence of postrenal AKI, they may identify the presence of calcium-containing stones that can cause obstructive disease. Renal ultrasonography is the screening test of choice to assess cortical thickness, differences in cortical and medullary density, the integrity of the collecting system, and kidney size.¹³³ Although pelvicalyceal dilation is usual in cases of urinary tract obstruction (98% sensitivity), dilation may not be observed in the volume-depleted patient during the initial 1 to 3 days after obstruction, when the collecting system is relatively noncompliant, or in patients with obstruction caused by ureteric encasement or infiltration (e.g., retroperitoneal fibrosis, neoplasia).¹³⁴

Alternatively, computed tomography (CT) may be used to visualize the kidneys and collecting system, although contrast administration should ideally be avoided in patients with AKI. Visualization of the collecting system may be suboptimal in the absence of contrast enhancement; however, unenhanced CT scans are useful for the identification of obstructing ureteral stones.^{135,136} Ultrasonography and CT have essentially replaced the use of IV pyelography, which now has little or no role in the evaluation of AKI. Cystoscopic retrograde or percutaneous antegrade pyelography are useful tests for the precise localization of the site of obstruction and can be combined with placement of ureteral stents or percutaneous nephrostomy tubes to allow therapeutic decompression of

the urinary tract. Radionuclide scans have been proposed as useful for assessing renal blood flow, glomerular filtration, tubule function, and infiltration by inflammatory cells in AKI; however, these tests lack specificity and yield conflicting or poor results in controlled studies.^{137,138} Magnetic resonance angiography (MRA) of the kidneys is extremely useful for detecting renal artery stenosis and has been used in the evaluation of acute renovascular crises.¹³⁹ However, given the association of gadolinium-based contrast administration with the development of nephrogenic systemic fibrosis, contrast-enhanced MRA is contraindicated in most patients with AKI.^{140,141} Doppler ultrasonography and spiral CT are also useful in patients with suspected vascular obstruction; however, contrast angiography remains the gold standard for definitive diagnosis.

KIDNEY BIOPSY

Kidney biopsy in AKI is usually reserved for patients in whom prerenal and postrenal AKI have been excluded, and the cause of intrinsic AKI is unclear.^{142,143} Kidney biopsy is particularly useful when clinical assessment, urinalysis, and laboratory investigation suggest diagnoses other than ischemic or nephrotoxic injury that may respond to specific therapy. Examples include anti-glomerular basement membrane disease and other forms of necrotizing glomerulonephritis, vasculitis, HUS and TTP, allergic interstitial nephritis, and myeloma cast nephropathy.

CAUSES OF ACUTE KIDNEY INJURY IN SPECIFIC CLINICAL SETTINGS

The differential diagnosis of AKI in several common clinical settings warrants special mention (Box 29.4).

ACUTE KIDNEY INJURY IN THE SETTING OF CANCER

There are several potential causes of AKI in the patient with cancer. Prerenal AKI is common in the setting of underlying malignancy and may be related to tumor- or chemotherapy-induced vomiting or diarrhea, reduced oral intake secondary to anorexia, the use of NSAIDs for pain management, and malignancy-associated hypercalcemia.^{144,145}

Intrinsic AKI can be triggered by a variety of chemotherapeutic agents. Cisplatin is the classic chemotherapeutic medication associated with AKI.^{146,147} The principal site of renal damage with cisplatin is the proximal tubule. The nephrotoxicity of cisplatin is dose-dependent, yet AKI can result from a single exposure. Electrolyte disturbances, including hypomagnesemia and hypokalemia, are common following cisplatin administration. Other platinum-containing chemotherapy agents, such as carboplatin and oxaliplatin, are less nephrotoxic than cisplatin but are not risk-free, particularly when high cumulative doses are administered. Ifosfamide, which has been used to treat germ cell tumors, sarcomas, other solid tumors, and occasionally lymphoma, is also associated with AKI in a dose-dependent fashion.^{148–150} Methotrexate nephrotoxicity occurs following IV administration of high doses ($>1 \text{ g/m}^2$), primarily as the result of precipitation of the drug and metabolites in the tubular

lumen.^{71,151,152} Risk factors for methotrexate nephrotoxicity include volume depletion and the presence of acidic urine. Direct tubular toxicity may also contribute to the development of AKI. Chemotherapeutic agents targeting vascular endothelial growth factor (VEGF) or the VEGF receptor, such as bevacizumab, and the tyrosine kinase inhibitor sunitinib, are associated with hypertension, proteinuria, thrombotic microangiopathy, and AKI.^{153,154} Checkpoint inhibitors, such as nivolumab, may cause a variety of immune-related adverse events, including AIN.^{155,156}

Renal parenchymal invasion by solid and hematologic cancers has been reported in 5% to 10% of autopsy studies but is an uncommon cause of AKI.^{157,158} Infiltration of leukemic cells into the renal parenchyma can precipitate AKI and typically presents with hematuria, proteinuria, and enlarged kidneys on ultrasound imaging. Prompt diagnosis is important because the AKI may respond to chemotherapeutic intervention.

The tumor lysis syndrome, which is associated with hyperuricemia, hyperphosphatemia, and hypocalcemia, is a well-recognized cause of AKI in patients with cancer.^{159,160} Tumor lysis syndrome occurs most commonly following the initiation of chemotherapy for patients with poorly differentiated, rapidly growing lymphoproliferative malignancies (e.g., Burkitt lymphoma, acute lymphoblastic or promyelocytic leukemia); however, this can occur spontaneously and in the setting of certain solid tumors that are highly sensitive to radiation and/or chemotherapy (e.g., testicular carcinoma). The Cairo-Bishop criteria, which include both laboratory and clinical criteria, have been used to provide a standard definition for the diagnosis of tumor lysis syndrome¹⁶⁰ (Box 29.5). AKI associated with the tumor lysis syndrome is thought to be triggered by direct tubular injury and luminal obstruction by uric acid and calcium phosphate crystals. Prophylactic therapy with aggressive volume administration and either xanthine oxidase inhibitors to inhibit uric acid synthesis or recombinant uricase to convert uric acid to allantoin has markedly reduced the incidence of this form of AKI.^{161–164} Less common causes of AKI include tumor-associated glomerulonephritis and thrombotic microangiopathy induced by medications or irradiation. Chemotherapy-associated thrombotic microangiopathy is a well-recognized complication of several agents, including mitomycin C and gemcitabine.^{165–167}

AKI is a common complication of multiple myeloma.^{75,168} Causes of AKI in this setting include intravascular volume depletion, myeloma cast nephropathy, sepsis, hypercalcemia, ATN induced by drugs or tumor lysis during therapy, cryoglobulinemia, hyperviscosity syndrome, and plasma cell infiltration. Multiple myeloma may also result in impaired kidney function as the result of amyloidosis or light chain deposition disease; however, these usually present with proteinuria and a more subacute decline in kidney function. Myeloma cast nephropathy results from the binding of filtered immunoglobulin Bence-Jones proteins to Tamm-Horsfall glycoprotein, forming casts that obstruct the tubular lumen. Higher excretion rates of free light chains, volume depletion, and hypercalcemia are associated with higher risks for the development of myeloma cast nephropathy. Prompt treatment to lower the free light chain burden may result in recovery of kidney function. Studies of the effectiveness of plasmapheresis in the treatment of myeloma cast nephropathy have yielded conflicting results.^{169–172} The use

Box 29.4 Major Causes of Acute Kidney Injury in Specific Clinical Settings**AKI in the Cancer Patient**

Prerenal azotemia

Hypovolemia (e.g., poor intake, vomiting, diarrhea)

Intrinsic AKI

Exogenous nephrotoxins—chemotherapy, antibiotics, contrast agents

Endogenous toxins—hyperuricemia, hypercalcemia, tumor lysis, paraproteins

Other—radiation, HUS, TTP, glomerulonephritis, amyloid, malignant infiltration

Postrenal AKI

Ureteric or bladder neck obstruction

AKI After Cardiac Surgery

Prerenal azotemia

Hypovolemia (surgical losses, diuretics), cardiac failure, vasodilators

Intrinsic AKI

Ischemic ATN (even in absence of hypotension)

Atheroembolic disease after aortic manipulation, intraaortic balloon pump

Pre- or perioperative administration of contrast agent

Allergic interstitial nephritis induced by perioperative antibiotics

Postrenal AKI

Obstructed urinary catheter, exacerbation of voiding dysfunction

AKI in Pregnancy

Prerenal azotemia—acute fatty liver of pregnancy with fulminate hepatic failure

Intrinsic AKI

Preeclampsia or eclampsia

Postpartum HUS, TTP

HELLP syndrome

Ischemia—postpartum hemorrhage, abruptio placentae, amniotic fluid embolus

Direct toxicity of illegal abortifacients

Postrenal AKI—obstruction with pyelonephritis

AKI After Solid Organ or Bone Marrow Transplantation (BMT)

Prerenal azotemia

Intravascular volume depletion (e.g., diuretic therapy)

Vasoactive drugs (e.g., calcineurin inhibitors, amphotericin B)

Hepatorenal syndrome, venoocclusive disease of liver (BMT)

Intrinsic AKI

Postoperative ischemic ATN (even in absence of hypotension)

Sepsis

Exogenous nephrotoxins: aminoglycosides, amphotericin B, radiocontrast

HUS, TTP (e.g., cyclosporine or myeloablative radiotherapy-related)

Allergic tubulointerstitial nephritis

Postrenal AKI

Obstructed urinary catheter

AKI and Pulmonary Disease (Pulmonary-Renal Syndrome)

Prerenal azotemia—diminished cardiac output complicating pulmonary embolism, severe pulmonary hypertension, or positive pressure mechanical ventilation

Intrinsic AKI

Vasculitis—Goodpasture syndrome, ANCA-associated vasculitis, SLE, Churg-Strauss syndrome, polyarteritis nodosa, cryoglobulinemia, right-sided endocarditis, lymphomatoid granulomatosis, sarcoidosis, scleroderma

Toxins—ingestion of paraquat or diquat

Infections—Legionnaires' disease, *Mycoplasma* infection, tuberculosis, disseminated viral or fungal infection

AKI from any cause with hypervolemia and pulmonary edema

Lung cancer with hypercalcemia, tumor lysis, or glomerulonephritis

AKI and Liver Disease

Prerenal azotemia

Reduced true (gastrointestinal [GI] hemorrhage, GI losses from lactulose, diuretics, large-volume paracentesis) circulatory volume or effective (hypoalbuminemia, splanchnic vasodilation)

Hepatorenal syndrome type 1 or 2

Tense ascites with abdominal compartment syndrome

Intrinsic AKI

Ischemic (severe hypoperfusion; see previously) or direct nephrotoxicity and hepatotoxicity of drugs or toxins (e.g., carbon tetrachloride, acetaminophen, tetracyclines, methoxyflurane)

Tubulointerstitial nephritis + hepatitis caused by drugs (e.g., sulfonamides, rifampin, phenytoin, allopurinol, phenindione), infections (leptospirosis, brucellosis, Epstein-Barr virus, cytomegalovirus), malignant infiltration (leukemia, lymphoma) or sarcoidosis

Glomerulonephritis or vasculitis (e.g., polyarteritis nodosa, ANCA-associated GN, cryoglobulinemia, systemic lupus erythematosus, postinfectious hepatitis or liver abscess)

AKI and Nephrotic Syndrome

Prerenal azotemia

Intravascular volume depletion (diuretic therapy, hypoalbuminemia)

Intrinsic AKI

Manifestation of primary glomerular disease

Collapsing glomerulopathy (e.g., HIV, pamidronate)

Associated ATN (older hypertensive men)

Associated interstitial nephritis (NSAIDs, rifampin, interferon alfa)

Other—amyloid or light chain deposition disease, renal vein thrombosis, severe interstitial edema

AKI, Acute kidney injury; ANCA, antineutrophil cytoplasmic antibody; ATN, acute tubular necrosis; GN, glomerulonephritis; HELLP, hemolysis, elevated liver enzymes, low platelets; NSAIDs, nonsteroidal antiinflammatory drugs; SLE, systemic lupus erythematosus.

of dialysis membranes that are permeable to light chains and other proteins with molecular weights lower than albumin (high-cutoff membranes) has also been proposed as a potential therapeutic strategy; however, data from clinical trials evaluating the efficacy of this strategy are similarly conflicting.^{173–175}

ACUTE KIDNEY INJURY IN PREGNANCY

In the industrialized world, the incidence of dialysis-requiring AKI in the setting of pregnancy is approximately 1 in 20,000 births.^{176,177} The marked decline in this complication over the past 50 years is a result of improved prenatal care and

Box 29.5 Cairo-Bishop Definition of Tumor Lysis Syndrome

Diagnosis of Laboratory Tumor Lysis Syndrome

Requires at least two of the following criteria achieved in the same 24-hour interval from 3 days before to 7 days after chemotherapy initiation:

- Uric acid ≥ 8.0 mg/dL or $\geq 25\%$ increase from baseline
- Potassium ≥ 6.0 mmol/L or $\geq 25\%$ increase from baseline
- Phosphorus ≥ 4.6 mg/dL (≥ 6.5 mg/dL in children) or $\geq 25\%$ increase from baseline
- Calcium ≤ 7.0 mg/dL or $\geq 25\%$ decrease from baseline

Diagnosis of Clinical Tumor Lysis Syndrome

Laboratory tumor lysis syndrome plus at least one of the following:

- Serum creatinine ≥ 1.5 times the age-adjusted upper limit of normal
- Cardiac arrhythmia/sudden death
- Seizure

Modified from Cairo MS, Bishop M. Tumour lysis syndrome: new therapeutic strategies and classification. *Br J Haematol.* 2004;127:3–11.

advancements in obstetrics practice. In early pregnancy, ATN induced by nephrotoxic abortifacients remains a relatively common cause of AKI in developing countries, but is rare in the developed world. Ischemic ATN, severe toxemia of pregnancy, and postpartum HUS and TTP are the most common causes of AKI in later term pregnancy.^{176,178,179} Ischemic ATN is usually precipitated by placental abruption or postpartum hemorrhage and, less commonly, by amniotic fluid embolism or sepsis. Glomerular filtration is usually normal in mild or moderate preeclampsia; however, AKI may complicate severe preeclampsia.^{179,180} In this setting, AKI is typically transient and found in association with intrarenal vasospasm, marked hypertension, and neurologic abnormalities.

A variant of preeclampsia, the HELLP syndrome (*hemolysis, elevated liver enzymes, low platelets*), is characterized by a benign initial course that can rapidly deteriorate with the development of thrombotic microangiopathy with hemolysis, coagulation abnormalities, derangement in hepatic function, and AKI.^{180–182} Immediate delivery of the fetus is indicated in this setting. Thrombotic microangiopathy can also develop in the postpartum setting and typically occurs in patients who have had a normal pregnancy.¹⁸³ Postpartum thrombotic microangiopathy is characterized by thrombocytopenia, microangiopathic anemia, and normal prothrombin and partial thromboplastin times and frequently results in long-term impairment of renal function.

Acute fatty liver of pregnancy (AFLP) occurs in approximately 1 in 7000 pregnancies and is associated with AKI, likely as a result of intrarenal vasoconstriction, as occurs in the hepatorenal syndrome. Although the exact origin of AFLP is unknown, the incidence is increased in women who carry a fetus with a defect in fatty acid oxidation and who are themselves carriers of a genetic mutation that compromises

intramitochondrial fatty acid oxidation.¹⁸² Acute bilateral pyelonephritis may also precipitate AKI in pregnancy and should be obvious from the patient's presentation (fever, flank pain), findings on urinalysis (bacteria, leukocytes), and laboratory tests (leukocytosis, elevated serum creatinine level).^{178,181,184,185} The diagnosis of postrenal AKI in the pregnant patient is particularly challenging due to the physiologic dilation of the collecting system that normally occurs in the second and third trimesters. As a result, determining the presence of abnormal findings on renal ultrasound is more difficult.

ACUTE KIDNEY INJURY IN THE SETTING OF CARDIAC SURGERY

An acute deterioration in kidney function is a relatively common complication following cardiac surgery, with an incidence of 7.7% to 42%, depending on the criteria used to define AKI.^{186–190} AKI requiring dialytic support occurs in up to 5% of patients following cardiac surgery.^{186–190} AKI in the perioperative period is usually attributed to prerenal azotemia associated with decreased cardiac function or to ATN. Risk factors for cardiac surgery–associated AKI can be broadly categorized into presurgical patient-related factors, surgical factors, and postoperative events. The principal patient-related risk factors include underlying CKD, advanced age, left ventricular dysfunction, previous myocardial revascularization, diabetes mellitus, and peripheral vascular disease.^{189–192} Operative factors include the need for emergent surgery, prolonged time on cardiopulmonary bypass, insertion of an intraaortic balloon pump, performance of concomitant valvular surgery, and redo coronary artery bypass grafting (CABG). Several studies have compared the incidence of AKI following on-pump versus off-pump CABG, with some data suggesting that off-pump CABG is associated with a lower incidence of AKI.^{193–197} Postoperative factors associated with an increased risk for AKI include reduced cardiac output, bleeding, vasodilatory shock, and the overzealous use of diuretics and afterload reducing agents.

Additional potential causes of AKI following CABG include the administration of iodinated contrast media in the pre-, peri-, and/or postoperative period, antibiotic-associated acute interstitial nephritis, and atheroembolic disease.¹⁹⁸ Whereas prerenal azotemia and ATN typically occur within days of the surgical procedure, atheroembolic AKI may take longer to develop and can be distinguished by the characteristic clinical features of livedo reticularis, cyanosis, and gangrenous digital lesions, as well as the findings of eosinophilia, eosinophiluria, and hypocomplementemia.

ACUTE KIDNEY INJURY AFTER SOLID ORGAN OR BONE MARROW TRANSPLANTATION

Nonrenal solid organ transplant recipients have a particularly high risk of AKI from cardiopulmonary and hepatic failure, sepsis, and the nephrotoxic effects of antimicrobial and immunosuppressive agents. In a large retrospective multicenter study, 25% of all nonrenal solid organ transplant recipients developed AKI, with 8% requiring RRT.¹⁹⁹ The development of AKI requiring dialysis was associated with a 9- to 12-fold increase in mortality. AKI developed in 35% of heart transplant recipients and in 15% of lung transplant

recipients. As many as 30% of liver transplant recipients develop AKI, many of whom had CKD prior to transplantation.^{200,201} There are conflicting data as to whether impaired kidney function pretransplantation predicts outcomes in patients undergoing orthotopic liver transplantation; however, patients with impaired kidney function preoperatively have longer hospital and ICU stays and are more likely to need dialysis compared with patients with intact preoperative kidney function.^{202–204}

AKI is a well-recognized complication of hematopoietic cell transplantation.^{144,205,206} The three types of hematopoietic cell transplantation are myeloablative autologous, myeloablative allogeneic, and nonmyeloablative allogeneic, and the incidence, severity, and outcomes of AKI following these forms of hematopoietic cell transplantation vary considerably.^{205,207,208} In a study of 272 patients who underwent myeloablative hematopoietic cell transplantation (predominantly allogeneic), 53% developed AKI and 24% required dialysis.²⁰⁹ Of patients with dialysis requiring AKI, the mortality rate was 84%. One study has found an incidence of severe AKI in this patient population of 73%.²¹⁰

AKI following nonmyeloablative allogeneic hematopoietic cell transplantation is less common.^{210,211} A study of 253 patients demonstrated an incidence of AKI of 40% within 3 months of hematopoietic cell transplantation, with just 4.4% of patients requiring dialysis.²¹¹ The incidence of AKI following myeloablative autologous hematopoietic cell transplantation is considerably lower.^{212,213} A study of 173 patients following autologous hematopoietic cell transplantation reported an incidence of AKI of 21%, with 5% of patients requiring dialysis.²¹³ The absence of graft-versus-host disease and more rapid engraftment likely account for the lower incidence of AKI in this setting. Causes of hematopoietic cell transplantation-associated AKI include hypovolemia, sepsis, tumor lysis syndrome, direct tubular toxicity from cytoreductive therapy, thrombotic microangiopathy, graft-versus-host disease, antibiotics, immunosuppressive agents, and hepatic venoocclusive disease (VOD).

VOD results from acute radiochemotherapy-induced endothelial cell injury of hepatic venules.^{209,214–216} This condition usually occurs in conditioning regimens that include total body irradiation and cyclophosphamide and/or busulfan and in the setting of myeloablative allogeneic hematopoietic cell transplantation. The syndrome is characterized clinically by profound jaundice and avid salt retention, with edema and ascites within the first month after engraftment and the subsequent development of AKI. Oliguric AKI is common in moderate VOD and certain in severe cases. The mortality rate for patients with severe VOD approaches 100%.

BK virus is a human polyoma virus that is a common opportunistic infection in solid organ transplant recipients and in patients following hematopoietic cell transplantation.²¹⁷ Detectable BK viruria may be seen in as many as 50% of patients undergoing hematopoietic cell transplantation.²¹⁸ Reactivation of latent BK virus infection in immunosuppressed patients is associated with both hemorrhagic cystitis and renal involvement with tubular atrophy and fibrosis, with an inflammatory lymphocytic infiltrate with intranuclear BK virus inclusion bodies.²¹⁹ The diagnosis is suggested by rising viral titers in blood and/or urine; the mainstay of treatment is minimization of immunosuppression.

ACUTE KIDNEY INJURY ASSOCIATED WITH PULMONARY DISEASE

The coexistence of AKI and pulmonary disease (pulmonary-renal syndrome) typically suggests the presence of Goodpasture syndrome, antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis, or other vasculitides.^{220–222} The detection of antiglomerular basement membrane antibodies, antineutrophil cytoplasmic antibodies, or low serum complement concentrations can be helpful in differentiating among the various causes of pulmonary-renal syndrome, although the urgent need for definitive diagnosis and treatment may mandate lung or renal biopsy. Several toxic ingestions and infections may also precipitate simultaneous pulmonary and kidney injury that mimics vasculitis-associated pulmonary-renal syndrome. Furthermore, AKI of any cause may be complicated by secondary hypervolemia and pulmonary edema. Severe lung disease and ventilator support with increased intrathoracic pressure may compromise cardiac output and induce prerenal AKI.

ACUTE KIDNEY INJURY IN ASSOCIATION WITH LIVER DISEASE

The differential diagnosis of AKI in patients with liver disease is broad. Common causes of AKI in this setting include intravascular volume depletion, gastrointestinal bleeding, sepsis, and nephrotoxins. Most cases of AKI in advanced liver disease are due to prerenal azotemia, ATN, or HRS, and differentiating these conditions can be clinically challenging.^{223–225} Although a urine sodium concentration less than 20 mmol/L and fractional excretion rate of sodium less than 1% are typical of prerenal AKI and HRS, high-dose diuretics, which are commonly prescribed in patients with advanced liver disease, may lead to higher sodium excretion rates. Differentiating ATN from other forms of AKI is further confounded by the fact that bile-stained casts, which can be seen in prerenal AKI and HRS, have a similar appearance to the classic, muddy brown, granular casts of ATN.²²⁶ Kidney disease in patients with liver disease may also result from acute glomerular disease, including immunoglobulin A (IgA) nephropathy, hepatitis B virus-associated membranous nephropathy, and hepatitis C virus-associated membranoproliferative glomerulonephritis with cryoglobulinemia. Acetaminophen toxicity may cause nephrotoxic ATN in addition to being one of the most common causes of acute hepatotoxicity.

The term *hepatorenal syndrome* is typically reserved for a clinical syndrome marked by irreversible AKI that develops in patients with advanced cirrhosis, although it has also been described in the setting of fulminant viral and alcoholic hepatitis. HRS almost certainly represents the terminal stage of a state of hypoperfusion that begins early in the course of chronic liver disease. The precise pathophysiologic mechanisms underlying the hemodynamic alterations in HRS are incompletely understood. In the early stages of HRS, increased vascular capacitance as the result of splanchnic and systemic vasodilation is thought to trigger activation of the renin-angiotensin-aldosterone system (RAAS) and sympathetic nervous system.²²³ Renal perfusion is preserved in this stage by the local release of renal vasodilatory factors; however, these compensatory mechanisms are eventually overwhelmed, and progressive renal hypoperfusion ensues.

An inadequate increase in cardiac output relative to the fall in vascular resistance is also thought to contribute to the development of HRS.

Clinically, the presentation of HRS closely resembles that of prerenal AKI. However, unlike prerenal AKI, HRS does not improve with aggressive expansion of the intravascular space. Criteria for the diagnosis of HRS have undergone revision (Box 29.6).²²⁷ Previous criteria were based on an increase in the serum creatinine level to more than 1.5 mg/dL in the setting of cirrhosis with ascites,²²⁸ whereas the updated criteria have been harmonized with the creatinine component of the KDIGO AKI definition.²²⁷ Other criteria include failure of kidney function to improve after at least 2 days with diuretic withdrawal and volume expansion with albumin, the absence of shock or concurrent or recent treatment with nephrotoxic drugs, and the absence parenchymal kidney disease—defined by proteinuria more than 500 mg/day, hematuria (>50 RBC/high power field [hpf]), and/or abnormal renal ultrasonography.²²⁷

Two subtypes of HRS have been described. Under the prior criteria, type 1 HRS was characterized by a rapid onset of AKI defined by at least a doubling of the serum creatinine concentration to a level of at least 2.5 mg/dL or a reduction in glomerular filtration of 50% or more to a level less than 20 mL/min over a 2-week period.^{229,230} In the revised criteria, the diagnosis of type 1 HRS is based on meeting KDIGO criteria for stage 2 or higher AKI (i.e., doubling of serum creatinine level from baseline).²²⁷ Type 1 HRS typically

develops in hospitalized patients and may be precipitated by variceal bleeding, overly rapid diuresis, the performance of paracentesis or, most commonly, the development of spontaneous bacterial peritonitis. Other postulated triggers include infections, minor surgery, or the use of NSAIDs or other drugs. However, caution must be exerted in these cases to exclude reversible causes of AKI. Type 1 HRS is generally characterized by a fulminant course with oliguria, encephalopathy, marked hyperbilirubinemia, and death within 1 month of the clinical presentation. However, advances in the management of HRS (discussed later) have suggested that there may be a trend toward better survival in patients who respond to therapy.^{231,232} Type 2 HRS is typified by a more gradual decline in renal function that develops in the setting of diuretic resistant ascites and avid sodium retention. The prognosis of type 2 HRS is considerably better than that of type 1 HRS, with a reported median survival of 6 months and a 1-year survival rate as high as 30%.^{233,234} The development of a sudden deterioration in kidney function after a prolonged stable period may occur in patients with type 2 HRS, leading to outcomes similar to those of patients with type 1 HRS.

Definitive treatment of HRS is dependent on the recovery of hepatic function or successful liver transplantation. However, the use of vasoconstrictive agents combined with volume expansion with colloid has shown promise for improving kidney function.^{235–237} It is postulated that by reversing the splanchnic and peripheral vasodilation, more normal renal perfusion can be restored. Vasoconstrictive regimens that have been used include norepinephrine, combination therapy with midodrine and octreotide, and the vasopressin agonist terlipressin.^{235–241} Although vasoconstrictive therapy is associated with improvement in kidney function, and patients who respond have an improved prognosis, the use of vasoconstrictive therapy has not been shown to improve overall prognosis in patients with AKI, suggesting that survival remains limited by the underlying severity of the liver disease.

Box 29.6 Diagnostic Criteria for Hepatorenal Syndrome

- Cirrhosis with ascites
- Acute kidney injury (AKI) defined as:
 - ≥ 0.3 -mg/dL increase in serum creatinine over <48 hours
 - $\geq 50\%$ increase in serum creatinine level known or presumed to have occurred within prior 7 days
- No improvement of serum creatinine level (decrease to a level of ≤ 1.5 mg/dL) after at least 2 days of diuretic withdrawal and volume expansion with albumin (1 g/kg body weight per day to a maximum of 100 g/day)
- Absence of shock
- Absence of parenchymal kidney disease as indicated by:
 - Proteinuria >500 mg/day
 - Microhematuria (>50 red blood cells/high power field); and/or
 - Abnormal renal ultrasonography

Type 1 Hepatorenal Syndrome

Rapid progressive AKI defined based on increase in serum creatinine level by $>2\times$ baseline

Type 2 Hepatorenal Syndrome

Moderate renal dysfunction with a steady or slowly progressive course

Modified from Angeli P, Gines P, Wong F, et al. Diagnosis and management of acute kidney injury in patients with cirrhosis: revised consensus recommendations of the International Club of Ascites. *Gut*. 2015;64:531–537.

ACUTE KIDNEY INJURY AND THE NEPHROTIC SYNDROME

AKI in the context of the nephrotic syndrome presents a unique array of potential diagnoses. Epithelial injury, if severe, can trigger both nephrotic range proteinuria and acute or subacute kidney injury.^{242,243} The epithelial injury typically occurs as a manifestation of primary glomerular disease, such as collapsing glomerulopathy or crescentic membranous nephropathy. Less dramatic visceral epithelial cell injury, in combination with proximal tubular injury (e.g., panepithelial cell injury induced by NSAIDs or possible undiagnosed viral illness) or interstitial nephritis (e.g., rifampicin- or ampicillin-induced) can also present as AKI complicating the nephrotic syndrome.^{244–246} Massive excretion of light chain proteins in patients with multiple myeloma may also present in this fashion.^{247,248} ATN in association with nephrotic syndrome is seen in a subpopulation of older patients with minimal change disease, and in other patients with nephrosis and severe hypoalbuminemia, particularly with overzealous diuresis. In general, patients with AKI complicating the nephrotic syndrome have higher blood pressure and urinary protein excretion than patients without AKI.²⁴² The higher incidence of arteriosclerosis in biopsy samples from these

patients may point to preexisting hypertensive nephrosclerosis as a risk factor for the development of this complication. Renal vein thrombosis must always be considered in the differential diagnosis of the nephrotic syndrome and AKI, particularly in the pediatric population and in adults with membranous nephropathy in association with high-grade proteinuria and hypoalbuminemia.

COMPLICATIONS OF ACUTE KIDNEY INJURY

The acute loss of kidney function in AKI results in multiple derangements in fluid, electrolyte, and acid-base homeostasis and in hematologic, gastroenterologic, and immunologic function (Table 29.6).

POTASSIUM HOMEOSTASIS

Hyperkalemia is a common and potentially life-threatening complication of AKI.^{249,250} The serum K^+ level typically rises by 0.5 mmol/L/day in oligoanuric patients and reflects impaired excretion of K^+ derived from a patient's diet, the administration of K^+ -containing solutions and drugs administered as potassium salts, and the release of K^+ from the injured tubular epithelium. Hyperkalemia may be compounded by coexistent metabolic acidosis and/or hyperglycemia or other hyperosmolar states that promote K^+ efflux from cells. Hyperkalemia present at the time of diagnosis of AKI or the rapid development of severe hyperkalemia suggests massive tissue destruction, as might be seen with rhabdomyolysis, hemolysis, or tumor lysis.^{50,90,251} Hyperuricemia and hyperphosphatemia may accompany hyperkalemia in these settings. Mild hyperkalemia (<6.0 mmol/L) is usually asymptomatic. Higher levels are frequently associated with electrocardiographic abnormalities, including, peaked T waves, prolongation of the PR interval, flattening of P waves, widening of the QRS complex, and intraventricular conduction defects.^{252–254} These electrocardiographic findings may precede the onset of life-threatening cardiac arrhythmias, such as bradycardia, heart block, ventricular tachycardia, ventricular fibrillation, and

asystole. In addition, hyperkalemia may induce neuromuscular abnormalities, such as paresthesias, hyporeflexia, weakness, ascending flaccid paralysis, and respiratory failure.

Hypokalemia is unusual in AKI but may complicate nonoliguric ATN caused by aminoglycosides, cisplatin, or amphotericin B, presumably because of impaired K^+ reabsorption resulting from epithelial cell injury in the thick ascending limb of the loop of Henle.^{255,256}

ACID-BASE HOMEOSTASIS

Normal metabolism of dietary protein yields between 50 and 100 mmol/day of fixed nonvolatile acids (principally sulfuric and phosphoric acids) that are excreted by the kidneys to maintain acid-base homeostasis. Predictably, AKI is commonly complicated by metabolic acidosis, typically with a widening of the serum anion gap due to retention of phosphates, sulfates, and organic anions.²⁵⁷ Acidosis may be severe (daily fall in plasma HCO_3^- >2 mmol/L) when the generation of H^+ is increased by additional mechanisms (e.g., diabetic or fasting ketoacidosis, lactic acidosis complicating generalized tissue hypoperfusion, liver disease, or sepsis, and metabolism of ethylene glycol).^{70,225,258} In contrast, metabolic alkalosis is an infrequent finding, but may complicate overly aggressive correction of acidosis with HCO_3^- , overzealous use of combination loop and thiazide diuretics, or loss of gastric acid by vomiting or nasogastric aspiration.

MINERAL AND URIC ACID HOMEOSTASIS

Mild to moderate hyperphosphatemia (5–10 mg/dL) is a common consequence of AKI, and hyperphosphatemia may be severe (10–20 mg/dL) in highly catabolic patients or when AKI is associated with rapid cell death as in rhabdomyolysis, severe burns, hemolysis, or tumor lysis.^{259–262} Factors that potentially contribute to hypocalcemia include skeletal resistance to the actions of parathyroid hormone, reduced levels of 1,25-dihydroxyvitamin D, Ca^{2+} sequestration in injured tissues, such as muscle in the setting of rhabdomyolysis, and metastatic deposition of calcium phosphate salts in the setting of severe hyperphosphatemia.^{263–265}

Table 29.6 Common Complications of Acute Kidney Injury

Metabolic	Cardiovascular	Gastrointestinal	Neurologic	Hematologic	Infectious	Other
Hyperkalemia	Pulmonary edema	Nausea	Neuromuscular irritability	Anemia	Pneumonia	Hiccups
Metabolic acidosis	Arrhythmias	Vomiting	Asterexis	Bleeding	Septicemia	Elevated parathyroid hormone level
Hyponatremia	Pericarditis	Malnutrition	Seizures		Urinary tract infection	Low total triiodothyronine and thyroxine
Hypocalcemia	Pericardial effusion	Hemorrhage	Mental status changes			Normal thyroxine level
Hyperphosphatemia	Pulmonary embolism					
Hypermagnesemia	Hypertension					
Hyperuricemia	Myocardial infarction					

Hypocalcemia is usually asymptomatic, possibly because of the counterbalancing effects of acidosis on neuromuscular excitability. However, symptomatic hypocalcemia can occur in patients with rhabdomyolysis or acute pancreatitis or after treatment of acidosis with HCO_3^- .²⁶³ Clinical manifestations of hypocalcemia include perioral paresthesias, muscle cramps, seizures, hallucinations, and confusion, as well as prolongation of the QT interval and nonspecific T wave changes on the electrocardiogram (ECG). The Chvostek sign (contraction of facial muscles on tapping of the jaw over the facial nerve) and Trousseau sign (carpopedal spasm after occlusion of arterial blood supply to the arm for 3 minutes with a blood pressure cuff) are useful indicators of latent tetany in high-risk patients.

Mild asymptomatic hypermagnesemia is common in oliguric AKI and reflects the impaired excretion of ingested magnesium—dietary magnesium, magnesium-containing laxatives, or antacids.^{266,267} More significant hypermagnesemia is usually the result of overzealous parenteral magnesium administration, as in the management of AKI associated with preeclampsia. Hypomagnesemia occasionally complicates nonoliguric ATN associated with cisplatin or amphotericin B and, as with hypokalemia, likely reflects injury to the thick ascending limb of loop of Henle, a principal site for Mg^{2+} reabsorption.^{256,268,269} Hypomagnesemia is usually asymptomatic but may occasionally manifest as neuromuscular instability, cramps, seizures, cardiac arrhythmias, or resistant hypokalemia or hypocalcemia.^{266,270}

Uric acid is cleared from blood by glomerular filtration and secretion by proximal tubule cells, and asymptomatic hyperuricemia (12–15 mg/dL) is typical in established AKI. Higher levels suggest increased production of uric acid and may point to a diagnosis of acute urate nephropathy.^{271–273} The urinary uric acid-to-creatinine ratio on a random specimen has been proposed as a means to distinguish between hyperuricemia caused by overproduction and impaired excretion. In a small series of patients, this ratio was more than 1 in 5 patients with acute uric acid nephropathy and was less than 1 in 27 patients with acute kidney injury due to other causes.²⁷⁴ In a subsequent case series, elevations in the uric acid-to-creatinine ratio to values of more than 1 were described in other etiologies of AKI, most notably patients with infections who were markedly hypercatabolic.²⁷⁵

VOLUME OVERLOAD AND CARDIAC COMPLICATIONS

Extracellular volume overload is an almost inevitable consequence of diminished salt and water excretion in AKI and may present clinically as mild hypertension, increased jugular venous pressure, pulmonary vascular congestion, pleural effusion, ascites, peripheral edema, increased body weight, and life-threatening pulmonary edema. Hypervolemia may be particularly troublesome in patients receiving multiple IV medications, high volumes of enteral or parenteral nutrition, and/or excessive volumes of maintenance IV fluids. Moderate or severe hypertension is unusual in ATN and should suggest other diagnoses, such as hypertensive nephrosclerosis, glomerulonephritis, preeclampsia, renal artery stenosis, and other diseases of the renal vasculature.^{178,276–278} Excessive water ingestion or the administration of a hypotonic saline or dextrose solution can trigger hyponatremia, which,

if severe, may cause cerebral edema, seizures, and other neurologic abnormalities.²⁷⁹ Cardiac complications include arrhythmias and myocardial infarction. Although these events may reflect primary cardiac disease, abnormalities in myocardial contractility and excitability may be triggered or compounded by hypervolemia, acidosis, hyperkalemia, and other metabolic sequelae of AKI.²⁸⁰

HEMATOLOGIC COMPLICATIONS

Anemia develops rapidly in AKI and is usually multifactorial in origin. Contributing factors include inhibition of erythropoiesis, hemolysis, bleeding, hemodilution, and reduced RBC survival time.^{281–283} Prolongation of the bleeding time is also common, resulting from mild thrombocytopenia, platelet dysfunction, and clotting factor abnormalities (e.g., factor VIII dysfunction).

NUTRITIONAL AND GASTROINTESTINAL COMPLICATIONS

Malnutrition remains one of the most frustrating and troublesome complications of AKI. Most patients have net protein breakdown, which may exceed 200 g/day in catabolic patients.^{284–286} Malnutrition is usually multifactorial in origin and may reflect an inability to eat, loss of appetite, and/or inadequate nutritional support, the catabolic nature of the underlying medical disorder (e.g., sepsis, rhabdomyolysis, trauma), nutrient losses in drainage fluids or dialysate, and increased breakdown and reduced synthesis of muscle protein and increased hepatic gluconeogenesis, probably through the actions of toxins, hormones (e.g., glucagon, parathyroid hormone), or other substances (e.g., proteases) that accumulate in AKI.^{287–291} Nutrition may also be compromised by the high incidence of acute gastrointestinal hemorrhage, which complicates up to 15% of cases of AKI. Mild gastrointestinal bleeding is common (10%–30%) and is usually due to stress ulceration of gastric or small intestinal mucosa.^{292,293}

INFECTIOUS COMPLICATIONS

Infection is the most common and serious complication of AKI, occurring in 50% to 90% of cases and accounting for up to 75% of deaths.^{25,249,294–296} It is unclear whether this high incidence of infection is due to a defect in host immune responses or to repeated breaches of mucocutaneous barriers (e.g., IV cannulas, mechanical ventilation, bladder catheterization) resulting from therapeutic interventions.

OTHER SEQUELAE OF ACUTE KIDNEY INJURY

Protracted periods of severe AKI or short intervals of catabolic anuric AKI often lead to the development of the uremic syndrome. Clinical manifestations of the uremic syndrome, in addition to those already listed, include pericarditis, pericardial effusion, and cardiac tamponade; gastrointestinal complications such as anorexia, nausea, vomiting, and ileus; and neuropsychiatric disturbances, including lethargy, confusion, stupor, coma, agitation, psychosis, asterixis, myoclonus, hyperreflexia, restless legs syndrome, focal neurologic deficit, and/or seizures. The uremic toxin(s) responsible for this

syndrome has (have) yet to be defined. Candidate molecules include urea, other products of nitrogen metabolism such as guanidine compounds, products of bacterial metabolism such as aromatic amines and indoles, and other compounds that are inappropriately retained in the circulation in AKI or are underproduced, such as nitric oxide (NO).²⁹⁷

COMPLICATIONS DURING RECOVERY FROM ACUTE KIDNEY INJURY

A vigorous diuresis may complicate the recovery phase of AKI and may precipitate intravascular volume depletion and can result in delayed recovery of kidney function. This diuretic response probably reflects the combined effects of an osmotic diuresis induced by retained urea and other byproducts of protein metabolism, excretion of retained salt and water accumulated during AKI, and delayed recovery of tubular reabsorptive function relative to glomerular filtration leading to salt wasting.^{298–301} Hyponatremia may also complicate this recovery phase if free water losses are not replenished or are inappropriately replaced by relatively hypertonic saline solutions. Hypokalemia, hypomagnesemia, hypophosphatemia, and hypocalcemia are rarer metabolic complications during recovery from AKI. Mild transient hypercalcemia is relatively frequent during recovery and appears to be a consequence of delayed resolution of secondary hyperparathyroidism. In addition, hypercalcemia may complicate recovery from rhabdomyolysis because of mobilization of sequestered Ca^{2+} from injured muscle.³⁰²

MANAGEMENT OF ACUTE KIDNEY INJURY

The treatment of AKI varies considerably, based on its cause and clinical presentation. Evidence-based pharmacologic therapy to counteract pathophysiologic processes in the kidney and arrest renal parenchymal damage is not available for certain forms of AKI, notably ATN. In such cases, the management of AKI focuses on implementing interventions to prevent its development, when possible, providing supportive care to ameliorate derangements of fluid and electrolyte homeostasis, and instituting treatment to prevent and mitigate uremic complications (Table 29.7). In cases of severe AKI, RRT is often required. The ultimate goals of management are to prevent death, facilitate recovery of kidney function, and minimize the risk for de novo and/or progressive CKD.

MANAGEMENT OF PRERENAL ACUTE KIDNEY INJURY

INTRAVASCULAR VOLUME DEPLETION

Prerenal AKI is defined as hemodynamically mediated kidney dysfunction that is rapidly reversible following normalization of renal perfusion.³⁰³ Prevention of prerenal AKI from intravascular volume depletion involves the early recognition and treatment of conditions that involve the loss of extracellular fluid, including vomiting, diarrhea, excessive diuresis, and bleeding before underperfusion

Table 29.7 Supportive Management of Acute Kidney Injury

Management Issue	Treatment
Intravascular volume overload	Restriction of salt (<1–2 g/day) and water (<1 L/day) intake Diuretic therapy (if nonoliguric) Ultrafiltration
Hyponatremia	Restriction of oral and intravenous free water
Hyperkalemia	Calcium gluconate (10 mL of 10% solution over 5 min) if ECG changes present Insulin (10–20 Units) IV push + glucose (250 mL of 20%) IV over 30–60 min Albuterol (10–20 mg by nebulizer or MDI) Renal replacement therapy Loop diuretics (if nonoliguric) K^+ binding resin Discontinue K^+ supplements or K^+ -sparing diuretics
Metabolic acidosis	Restriction of dietary potassium Restriction of dietary protein Sodium bicarbonate (if $\text{HCO}_3^- < 15 \text{ mmol/L}$) Renal replacement therapy
Hyperphosphatemia	Restriction of dietary phosphate intake Phosphate binding agents (aluminum hydroxide, calcium carbonate, calcium acetate, sevelamer, lanthanum)
Hypocalcemia	Oral or intravenous replacement (if symptomatic or sodium bicarbonate to be administered)
Hypermagnesemia	Discontinue magnesium containing antacids
Nutrition	Caloric intake: 20–30 kcal/day Protein intake: Nondialysis-requiring—0.8–1.0 g/kg/day Dialysis-requiring—1.0–1.5 g/kg/day Continuous renal replacement therapy—up to 1.7 g/kg/day Enteral route of nutrition preferred
Drug dosage	Adjust all doses for GFR and renal replacement modality

ECG, Electrocardiographic; GFR, glomerular filtration rate; MDI, metered-dose inhaler.

of the kidneys occurs. In patients in whom intravascular volume depletion leads to prerenal AKI, treatment consists of restoration of a normal circulating blood volume. The optimal composition of administered fluids in patients with hypovolemic prerenal AKI depends on the source of fluid loss and associated electrolyte and acid-base disturbances. The initial management commonly consists of intravascular volume resuscitation with an isotonic crystalloid solution. Recent studies have demonstrated that balanced crystalloids reduce major adverse kidney events compared with isotonic saline in hospitalized patients.^{304,305} Red blood cell transfusion should be used for hemorrhagic hypovolemia when there is ongoing bleeding, particularly if the patient is hemodynamically unstable, or if the blood hemoglobin concentration is dangerously low.

The relative merits of colloid and crystalloid resuscitation fluids in the management of nonhemorrhagic renal, extrarenal, and third space fluid losses are controversial, with advocates for the use of colloids positing that they are more effective at restoring circulating blood volume due to greater retention in the intravascular compartment. However, randomized controlled trials (RCTs) and meta-analyses comparing colloid with crystalloid replacement for resuscitation in critically ill patients have not confirmed this theoretic benefit and demonstrated an increased need for RRT, and other adverse outcomes were associated with colloid formulations containing hydroxyethyl starch.^{306–313} In a meta-analysis of 55 trials involving 3504 patients randomly assigned to treatment with albumin or crystalloid, there was no evidence of improved outcomes, decreased mortality, or other complications associated with albumin administration.³¹⁴ These results were subsequently confirmed in a nearly 7000-patient multicenter RCT of fluid resuscitation in hypovolemic medical and surgical ICU patients, in which 28-day survival, development of single or multiple organ failure, and duration of hospitalization were similar in both groups.³¹⁵ Although specific data on the development of AKI were not described, the need for RRT was similar with saline compared with albumin resuscitation. However, in a post hoc analysis of patients with traumatic brain injury, albumin resuscitation was associated with increased mortality risk.³¹⁶

The use of synthetic colloid solutions has been proposed as an alternative to albumin administration; however, hydroxyethyl starch preparations have been associated with an increased risk of AKI. In a multicenter RCT comparing fluid resuscitation with hydroxyethyl starch with a 3% gelatin solution in 129 patients with sepsis, hydroxyethyl starch was associated with a more than twofold increased risk of AKI.³⁰⁹ A subsequent meta-analysis has confirmed the increased risk of AKI associated with hydroxyethyl starch across 34 studies that included 2604 individuals.³¹⁰ In an ensuing RCT that included 7000 critically ill patients who were assigned to receive 6% hydroxyethyl starch or isotonic saline, there was an approximately 20% increased risk of AKI treated with RRT with the use of hydroxyethyl starch.³¹³ Based on these data demonstrating no benefit and potential increased risk of AKI, along with the higher costs associated with colloid administration, their routine use for volume resuscitation in hypovolemia and sepsis is not advisable. In particular, hydroxyethyl starch solutions should be used very sparingly and, if used, there should be regular monitoring of kidney function. In such cases, the risk of hyperoncotic renal failure

should be minimized by the concomitant use of appropriate crystalloid solutions.^{3,309,310,312,313}

Experimental data have suggested that volume resuscitation with isotonic sodium chloride solutions, which contain supraphysiologic concentrations of chloride, may exacerbate renal vasoconstriction and diminish the GFR, as compared with isotonic crystalloid solutions with a lower chloride content.^{317–319} In healthy patients, magnetic resonance imaging (MRI) has demonstrated that the infusion of isotonic saline is associated with reduced renal blood flow velocity and renal cortical tissue perfusion as compared with administration of a reduced chloride isotonic crystalloid solution.³²⁰ In a subsequent open-label, sequential period study conducted in a single ICU, replacing the use of high-chloride IV solutions with fluids containing a lower chloride content was associated with a reduction in the incidence of KDIGO stage 3 AKI, from 14% to 8.4%, and in the use of RRT from 10% to 6.3%.³²¹ A subsequent meta-analysis that included 21 studies, 15 of which were small RCTs, found that the use of high-chloride IV fluid was associated with an increased risk of AKI, with no effect on mortality.³²² However, exclusion of heavily weighted studies in this analysis rendered the association of high-chloride fluid with AKI nonstatistically significant.

Two more recent large pragmatic, cluster-randomized clinical trials conducted in parallel at the same institution compared the administration of isotonic saline with balanced crystalloid solution (i.e., fluid with electrolyte composition that more closely resembles plasma).^{304,305} In the Saline Against Lactated Ringer's of Plasma-Lyte in the Emergency Department (SALT-ED) trial of over 13,000 noncritically ill patients, balanced crystalloid was associated with a decrease in the incidence of 30-day major adverse kidney events (i.e., death, new RRT, persistent kidney impairment defined by a >200% increase in serum creatinine level at the time of hospital discharge, within 30 days) compared with saline (4.7% vs. 5.6%; $P = .01$), but not with the primary outcome of hospital-free days (days alive after discharge to day 28).³⁰⁴ The Isotonic Solutions and Major Adverse Renal Events Trial (SMART), which included 15,802 critically ill patients, found that compared with isotonic saline, balanced crystalloids were associated with a decrease in 30-day major adverse kidney events (14.3% vs. 15.4%; $P = .04$).³⁰⁵ In the SALT-ED trial, the benefit in the composite outcome was predominantly due to a lower rate of persistent kidney impairment, whereas in SMART, 30-day mortality predominated. Collectively, these findings support a benefit of balanced crystalloid compared with isotonic sodium chloride, yet the benefits were not homogenous. Greatest benefit was present among the critically ill patients with sepsis and among non-critically ill patients who had a baseline serum creatinine level more than 1.5 mg/dL or hyperchloremia ($\text{Cl} > 110 \text{ mmol/L}$) on initial presentation. Although the balanced fluids were more physiologic with regard to their chloride content, they were hypotonic and associated with higher rates of hyponatremia.

The volume and electrolyte content of urinary and gastrointestinal losses, as well as patients' serum electrolyte and acid-base status, should be closely monitored to guide adjustments in the composition of the replacement fluids. Although the potassium content in gastric juices tends to be low, concomitant urinary potassium losses may be quite high as the result of metabolic alkalosis.

HEART FAILURE

The management of AKI in the setting of heart failure is dependent on the clinical setting and cause of the heart failure. In patients with heart failure in whom AKI has developed in the setting of excessive diuresis, withholding diuretics and administering cautious volume replacement may be sufficient to restore kidney function. In acute decompensated heart failure (ADHF), AKI may develop despite worsening volume overload; intensification of diuretic therapy is often required for treatment of pulmonary vascular congestion. Although diuretic therapy may exacerbate prerenal AKI, it can also result in improvement in kidney function via several postulated mechanisms:

1. Decreasing ventricular distention resulting in a shift from the descending limb to the ascending limb of the Starling curve and improvement in myocardial contractility
2. Decreasing venous congestion^{318,323–326}
3. Diminishing intraabdominal pressure³²⁷

Additional therapies for ADHF in the setting of AKI include inotropic support, vasodilators for afterload reduction, and mechanical support, including intraaortic balloon pumps and ventricular assist devices. The use of invasive hemodynamic monitoring in ADHF has been controversial; although it is often used to guide pharmacologic management, clinical data have not demonstrated improved renal outcomes when management is guided by pulmonary artery catheters.³²⁸ The role of isolated ultrafiltration in ADHF is also controversial. Although negative fluid balance can be achieved more readily using extracorporeal ultrafiltration than conventional diuretic therapy, studies have not demonstrated differences in kidney function or survival.^{329–331} In the Ultrafiltration Versus Intravenous Diuretics for Patients Hospitalized for Acute Decompensated Heart Failure (UNLOAD) trial, hypervolemic patients with heart failure who were randomized to isolated ultrafiltration had more rapid fluid loss and decreased rehospitalizations within 90 days as compared with patients randomized to diuretic therapy, with no differences in kidney function.³³⁰ In contrast, in the subsequent Cardiorenal Rescue Study in Acute Decompensated Heart Failure (CARRESS-HF) trial, ultrafiltration was inferior to diuretic therapy with respect to the bivariate endpoint of change in serum creatinine level and body weight 96 hours after enrollment ($P = .003$), owing primarily to worsening of kidney function in the ultrafiltration group.³³¹ Based on these data, extracorporeal ultrafiltration cannot be recommended for the primary management of patients with decompensated heart failure.

LIVER FAILURE AND HEPATORENAL SYNDROME

Although volume-responsive prerenal azotemia is common in patients with advanced liver disease, differentiation from HRS and intrinsic AKI may be difficult.²²³ Patients with liver failure are typically total body sodium overloaded with peripheral edema and ascites, but true hypovolemia or reduced effective systemic arterial blood volume is often an important contributory factor to the development AKI. The underlying pathophysiology of salt and water retention in cirrhosis involves multiple pathways. Portal hypertension leads directly to ascites formation, whereas splanchnic and peripheral vasodilation result in a state of relative arterial underfill-

ing, which activates neurohumoral vasoconstrictors that produce intrarenal vasoconstriction, salt and water retention, and decreased GFR.³³² Volume-responsive AKI may develop in the setting of excessive diuresis, increased gastrointestinal losses (often as the result of therapy for hepatic encephalopathy), rapid drainage of ascites, or spontaneous bacterial peritonitis. Worsening hepatic function is often associated with diuretic resistance and progressive or precipitous worsening of kidney function. It has been postulated that an inadequate increase in cardiac output in response to the fall in peripheral vascular resistance may be central to the development of the hepatorenal syndrome.³³³

Differentiation between volume-responsive prerenal AKI and the HRS is based on the clinical response to volume loading. The optimal fluid for volume expansion in this setting has been controversial. Most recent expert opinion has advocated the use of hyperoncotic (20% or 25%) albumin at a dose of 1 g/kg per day.^{228,334} However, there is an absence of rigorous data supporting this regimen as compared with volume expansion with isotonic crystalloid solutions. There are more data regarding the use of albumin infusion to prevent AKI in patients undergoing large-volume (>5 L) paracentesis^{234,335,336} and in the treatment of spontaneous bacterial peritonitis.³³⁷ In an RCT, patients undergoing paracentesis who received an infusion of approximately 10 g of albumin/L of drained ascites experienced less activation of the RAAS and a significantly lower rate of worsening kidney function than patients who did not receive albumin infusion.³³⁵ In a subsequent study, albumin infusion was superior to the administration of dextran or gelatin solutions in preventing AKI following large-volume paracentesis.³³⁶ Current recommendations are to infuse 6 to 8 g of albumin/L of ascites drained when the paracentesis volume exceeds 5 L. In an RCT comparing antibiotics alone with antibiotics plus albumin for the treatment of spontaneous bacterial peritonitis, infusion of 1.5 g/kg of albumin at the initiation of treatment and an additional 1 g/kg on the third day of treatment was associated with reduced rates of AKI and mortality,³³⁷ although the benefit appears to be restricted to patients in whom the serum creatinine level is more than 1 mg/dL, the blood urea nitrogen is more than 30 mg/dL or the total bilirubin is more than 4 mg/dL.³³⁸

Definitive therapy of hepatorenal syndrome requires restoration of hepatic function, usually achieved through liver transplantation.^{225,334} The role of peritoneovenous shunting (e.g., LeVeen and Denver shunts) in HRS has been inadequately studied. In a subset of 33 patients with HRS included in a randomized trial comparing peritoneovenous shunts to medical therapy, shunting was not associated with improved survival.³³⁹ These data need to be interpreted with caution due to the small sample size and because data on improvement in kidney function were not reported. In addition, as a result of poor long-term patency rates and high rates of complications, particularly encephalopathy, the use of peritoneovenous shunts has largely been supplanted by the use of a transjugular portosystemic shunt (TIPS). TIPS has been demonstrated to provide better control of ascites than sequential paracentesis^{340–343} and in one series, lower rates of HRS,³⁴¹ albeit with a higher risk of encephalopathy.³⁴⁴ In a small case series, TIPS was reported to be effective as primary therapy for hepatorenal syndrome,³⁴⁵ but has not been evaluated in a randomized trial.³³⁴

Pharmacologic therapy with vasoconstrictors, when combined with albumin infusion, has been associated with improvement in kidney function in patients with hepatorenal syndrome.^{334,346} Agents that have shown benefit include norepinephrine,³⁴⁷ the combination of octreotide and midodrine,^{231,348–350} and the V1 vasopressin receptor agonist terlipressin,^{235–237,351} although only terlipressin has been evaluated in RCTs. In metaanalyses of published trials, terlipressin was associated with a 3.5- to 4-fold increased odds of reversal of HRS.^{352,353} Treatment of HRS with terlipressin was also associated with a modest short-term reduction in mortality; however, longer term outcomes are primarily a function of the underlying liver disease rather than treatment of HRS.³⁵³ In addition, terlipressin was associated with a markedly increased risk of adverse cardiovascular events. More recently, the Reversal of Hepatorenal Syndrome Type 1 with Terlipressin (REVERSE) trial found similar rates of confirmed reversal of HRS with terlipressin and placebo (19.6% vs. 13.1%; $P = .22$), although the mean decrease in serum creatinine level was more pronounced with terlipressin (1.1 mg/dL vs. 0.6 mg/dL; $P = .001$).²³⁷ At present, terlipressin is not approved for use in the United States.

ABDOMINAL COMPARTMENT SYNDROME

Acute kidney injury can result from elevations in intra-abdominal pressure, resulting in a clinical presentation with similar features to prerenal AKI. The abdominal compartment syndrome is defined by an intraabdominal pressure 20 mm Hg or higher associated with dysfunction of one or more organ systems.³⁵⁴ However, intraabdominal pressures lower than 20 mm Hg may be associated with abdominal compartment syndrome, whereas values higher than this threshold do not universally lead to the abdominal compartment syndrome.^{355–358} Abdominal compartment syndrome typically develops in critically ill patients, usually in the setting of trauma with abdominal hemorrhage, abdominal surgery, massive fluid resuscitation, liver transplantation, and gastrointestinal conditions, including peritonitis and pancreatitis. Mechanisms underlying the development of AKI in abdominal compartment syndrome are believed to involve renal vein compression, reduced cardiac output, and renal arterial constriction from sympathetic nervous system and RAAS activation.^{359–361} Oliguria, which can lead to anuria, often develops and, as is true for other forms of AKI associated with impaired renal perfusion, urine sodium concentration is commonly reduced.

The diagnosis of abdominal compartment syndrome, which should be suspected in patients with acute abdominal distention and/or rapidly accumulating ascites or abdominal trauma, can be made by simple transduction of the bladder pressure.^{59,354,355} Treatment is prompt abdominal decompression; if ascites is present, decompression may be achieved by performing large-volume paracentesis and in patients with severe ileus or colonic distention, bowel decompression may be sufficient; however, surgical laparotomy is often required for definitive therapy.

MANAGEMENT OF POSTRENAL ACUTE KIDNEY INJURY

The principle underlying the management of postrenal AKI is the prompt relief of urinary tract obstruction. This topic

is reviewed extensively in Chapter 37. Urethral or bladder neck obstruction may be relieved with the placement of a transurethral or suprapubic bladder catheter. Similarly, ureteric obstruction may be acutely relieved by placement of percutaneous nephrostomy tubes or by cystoscopically placed ureteral stents. Following the initial relief of obstruction, most patients experience a physiologic diuresis that resolves after several days as the result of the excretion of volume and solutes retained during the period of renal obstruction. However, approximately 5% of patients may have a more prolonged diuretic phase following relief of obstruction because of delayed recovery of tubule function relative to the GFR, resulting in a salt-wasting syndrome, which may require IV fluid replacement to maintain blood pressure.^{300,301,362} Following initial relief of obstruction, urologic evaluation is required for definitive evaluation and management of the underlying cause of obstruction.

PREVENTION OF INTRINSIC ACUTE KIDNEY INJURY

GENERAL PRINCIPLES

Strategies to prevent intrinsic AKI vary based on the specific cause of the kidney injury. Optimization of cardiovascular function and restoration of intravascular volume status are key interventions to minimize the risk that prerenal AKI evolves into ischemic ATN. There is compelling evidence that aggressive intravascular volume expansion reduces the incidence of ATN after major surgery or trauma, burns, and cholera.^{53,306,363,364} AKI due to sepsis is common and is associated with mortality rates as high as 80%.^{35,296,365} The role of early goal-directed therapy (EGDT) using resuscitation to defined hemodynamic targets (mean arterial pressure [MAP] >65 mm Hg; central venous pressure [CVP], 10–12 mm Hg; urine output >0.5 mL/kg per hour; $ScvO_2 >70\%$), using a combination of crystalloid solutions, red cell transfusion, and vasopressors guided by invasive hemodynamic monitoring in improving overall outcomes and decreasing the risk of AKI, has been controversial. In a seminal single-center RCT, EGDT resulted in a significant reduction in overall organ dysfunction and mortality in patients presenting with severe sepsis or septic shock, although specific data on the incidence of AKI were not reported.³⁶⁶ However, EGDT was not associated with a reduction in dialysis-requiring AKI in the Protocolized Care for Early Septic Shock (ProCESS), Australasian Resuscitation in Sepsis Evaluation (ARISE), or Protocolised Management in Sepsis (PRoMISe) trials or in a patient-level meta-analysis that combined data from these trials (Fig. 29.1).^{367–370}

Although the benefits of EGDT were not confirmed in these three later trials, early recognition of sepsis, prompt initiation of antibiotic therapy, and rapid volume resuscitation and hemodynamic stabilization were found to improve outcomes and are likely to minimize the risk of AKI.³⁷¹ The role of maintenance of normoglycemia in critically ill patients in minimizing the risk of AKI has also been controversial. Two single-center RCTs that used intensive insulin management to maintain blood glucose levels of 80 to 110 mg/dL, as compared with conventional management maintaining the glucose concentration between 180 and 220 mg/dL, each resulted in decreased rates of AKI, defined either on the basis of change in the serum

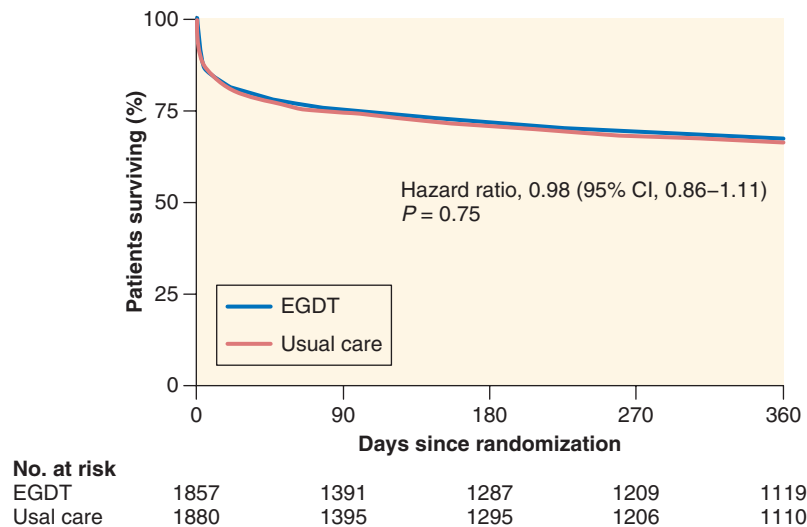


Fig. 29.1 One-year survival comparing early goal-directed therapy (EGDT) with usual care in patients with sepsis. Shown are Kaplan-Meier survival curves comparing EGDT with usual care in patients with sepsis from pooled patient-level data from the ProCESS, ARISE, and ProMISe trials. Renal replacement therapy was required in 11.0% of patients who were randomized to EGDT as compared with 10.6% of patients randomized to usual care (odds ratio, 1.02; 95% confidence interval, 0.81–1.28; $P = .88$). (From PRISM Investigators; Rowan KM, Angus DC, Bailey, et al. Goal-directed therapy for septic shock—a patient-level metaanalysis. *N Engl J Med*. 2017;376:2223–2234).

creatinine level or on the need for RRT.^{372–374} However, the benefits of tight glycemic control were not confirmed in the Normoglycemia in Intensive Care Evaluation–Survival Using Glucose Algorithm Regulation (NICE-SUGAR) trial, a 6104-patient multicenter trial that compared intensive therapy to achieve a target glucose of approximately 80 to 110 mg/dL with more conventional therapy designed to maintain the blood glucose level less than 180 mg/dL.³⁷⁵ In the NICE-SUGAR trial, intensive glycemic control was associated with an increased risk of hypoglycemia, an increased mortality risk (27.5% vs. 24.9%; $P = .02$), and no reduction in the need for RRT.

In surgical patients, avoidance of hypotension has been associated with a decreased risk of AKI. In a retrospective analysis of over 33,000 patients who underwent noncardiac surgery, episodes of intraoperative hypotension with a mean arterial blood pressure less than 55 mm Hg were associated with a marked increase in the probability of AKI.³⁷⁶ The adjusted odds of AKI with intraoperative hypotension increased with the duration of hypotension, from an odds ratio of 1.2 with less than 10 minutes of hypotension to 1.3 with 10 to 20 minutes of hypotension and to 1.5 with more than 20 minutes of hypotension. Volume overload is typical after surgical procedures, and small trials have suggested better patient outcomes following abdominal surgery using a restrictive fluid management strategy. However, a large RCT comparing restrictive and liberal fluid management strategies in patients undergoing major abdominal surgery at increased risk of complications found no benefit with regard to the primary outcome of 1-year disability-free survival, but found a higher rate of AKI associated with the restrictive fluid strategy (8.6% vs. 5.0%; $P < .001$).³⁷⁷

Intravascular volume depletion has been identified as a risk factor for ATN resulting from iodinated contrast, rhabdomyolysis, hemolysis, cisplatin, amphotericin B, multiple myeloma, aminoglycosides, and other nephrotoxins, crystal-associated

AKI related to acyclovir and acute urate nephropathy, and AKI stemming from hypercalcemia.^{74,75,160,169,251,273,364,378–380}

The restoration of adequate intravascular volume prevents the development of experimental and human ATN in many of these clinical settings. Avoidance of potentially nephrotoxic medications or insults in high-risk patients and settings is also important to reduce the risk for ATN. Specifically, among patients with advanced cardiac and/or liver disease in whom renal perfusion may be diminished, the use of selective or nonselective NSAIDs that inhibit the production of vasodilatory prostaglandins may exacerbate intrarenal vasoconstriction and precipitate AKI.^{381–385} Diuretics, NSAIDs (including selective cyclooxygenase-2 [COX-2] inhibitors), ACE inhibitors, ARBs, and other inhibitors of the RAAS should be used with caution in patients with suspected absolute or effective intravascular volume depletion or in patients with renovascular disease because these agents may convert reversible prerenal AKI to ischemic ATN. The combined use of agents that block the RAAS, diuretics, and NSAIDs has been identified as a risk factor for AKI, particularly among patients with heart failure, liver failure, or other conditions with reduced baseline renal perfusion.^{386,387}

Careful monitoring of circulating drug levels appears to reduce the incidence of AKI associated with aminoglycoside antibiotics and calcineurin inhibitors.^{388–390} The observation that the antimicrobial efficacy of aminoglycosides persists in tissues, even after the drug has been cleared from the circulation (postantibiotic killing), has led to the use of once-daily dosing with these agents. Dosing regimens that provide higher peak drug levels but less frequent administration appear to provide comparable antimicrobial activity and less nephrotoxicity than older, conventional dosing regimens.^{390–393} Nephrotoxicity of drugs may also be reduced through changes in formulation. For example, the use of lipid-encapsulated formulations of amphotericin B may decrease the risk of amphotericin-induced AKI.³⁹⁴

CONTRAST-ASSOCIATED ACUTE KIDNEY INJURY

Contrast-associated AKI (CA-AKI) has historically been defined by and commonly manifests clinically as small absolute (>0.5 mg/dL) and/or relative ($>25\%$) increases in serum creatinine levels that occur within 2 to 4 days following iodinated contrast administration. Although severe AKI is relatively uncommon due to iodinated contrast alone, the incidence of CA-AKI defined by relatively minor increments in the serum creatinine level has been shown in past studies to occur in up to 15% or more of high-risk patients.³⁹⁵ A series of observational retrospective studies have questioned the true incidence of CA-AKI and challenged the concept that iodinated contrast is nephrotoxic.^{396–399} A meta-analysis of 13 nonrandomized studies that included 25,950 patients found a similar risk of AKI in patients who received iodinated contrast IV compared with patients who underwent radiographic procedures without intravascular contrast (relative risk [RR], 0.79; 95% confidence interval [CI], 0.62–1.02).³⁹⁷ Similarly, an observational study of over 29 million patient hospitalizations found that the risk for AKI among patients who received intravascular iodinated contrast during their hospitalization was comparable to the risk observed among patients who had not received contrast (RR, 0.93; 95% CI, 0.88–0.97).⁴⁰⁰ Although the results of these retrospective analyses are adjusted for medical comorbidity and other risk factors for renal injury, such adjustment cannot fully account for all factors that influence the decision to administer contrast, rendering it likely that patients in these studies who did not receive contrast may have been at higher risk for AKI than patients who received contrast. It is therefore prudent to continue to consider intravascular iodinated contrast as potentially nephrotoxic and to implement evidence-based preventive care in patients at risk for CA-AKI (Table 29.8).

Past studies have demonstrated that the administration of IV fluids to high-risk patients prior to and following exposure to intravascular iodinated contrast diminished the risk for CA-AKI, although the optimal regimen for fluid administration is unknown.^{3,401,402} In a small clinical trial that was stopped

early due to safety concerns, periprocedural IV isotonic saline was associated with a markedly lower rate of AKI after contrast exposure than oral fluid administration.⁴⁰¹ In a larger randomized trial, isotonic IV saline significantly reduced the incidence of CA-AKI following coronary angiography compared with half-normal IV saline, with a particular benefit noted in diabetic patients and those receiving large volumes of contrast.⁴⁰²

A recent study has challenged the principle that IV fluid administration reduces the risk for CA-AKI.⁴⁰³ This noninferiority clinical trial randomized 660 patients with baseline CKD undergoing procedures with iodinated contrast to receive isotonic saline IV or no IV fluids and found a similar rate of CA-AKI (2.7% with saline vs. 2.6% without fluid), concluding that withholding IV fluid was not inferior to administering IV saline with regard to the prevention of small increments in the serum creatinine level. However, the patient population in this study was at relatively low overall risk for CA-AKI because more than 50% of participants received IV rather than intraarterial contrast, which is associated with lower rates of CA-AKI, and most had less advanced CKD (i.e., estimated GFR, 45–59 mL/min/1.73 m²). Furthermore, this trial was powered based on the recruitment of 1300 patients, but enrolled 660 due to feasibility considerations. Thus, until additional clinical trials have demonstrated conclusively that isotonic IV fluid is ineffective, this treatment remains the standard of care for the prevention of CA-AKI.

A number of small clinical trials have compared the effects of isotonic sodium bicarbonate compared with isotonic saline for the prevention of CA-AKI.^{404–413} These studies were generally underpowered and yielded conflicting results. Subsequent meta-analyses have concluded that there is an overall benefit associated with bicarbonate administration with regard to AKI defined by small changes in the serum creatinine level, although there was no demonstrable benefit with regard to the need for dialysis.^{414–416} This led several clinical practice guidelines to recommend the administration of either IV isotonic sodium chloride or sodium bicarbonate to high-risk patients receiving iodinated contrast.^{3,417,418} However, the Prevention of Serious Adverse Events Following Angiography (PRESERVE) trial provided more definitive findings regarding the comparative effects of sodium bicarbonate and sodium chloride for the prevention of CA-AKI.⁴¹⁹ PRESERVE was a multinational randomized clinical trial that used a 2×2 factorial design to compare IV isotonic sodium bicarbonate with IV isotonic sodium chloride and oral *N*-acetylcysteine with placebo for the prevention of serious adverse outcomes and CA-AKI following angiographic procedures in 4993 patients with CKD. IV sodium bicarbonate did not reduce the incidence of a primary outcome comprised of 90-day death, need for dialysis, or persistent decline in kidney function (odds ratio [OR], 0.93; 95% CI, 0.72–1.22). Similarly, bicarbonate did not reduce the incidence of CA-AKI assessed 3 to 5 days postangiography (OR, 1.16; 95% CI, 0.96–1.41). Although the optimal rate and duration of IV saline administration is not known, it is reasonable to administer isotonic sodium chloride at a rate of 1 mL/kg per hour for 6 to 12 hours prior to and 6 to 12 hours following a contrast-enhanced procedure in at-risk hospitalized patients. For at-risk outpatients, an alternative regimen of 3 mL/kg over 1 hour prior to the procedure, followed by 6 mL/kg administered over 2 to 6 hours following the procedure, may be more feasible.

Table 29.8 Effectiveness of Preventive Interventions for Contrast-Associated Acute Kidney Injury

No Benefit	Unclear Benefit	Beneficial
Loop diuretics ^a	Statins	Isotonic intravenous fluids
Mannitol ^a	Natriuretic peptides	Low- or iso-osmolal contrast media
Dopamine ^a	Theophylline, aminophylline	
Fenoldopam ^a	Ascorbic acid	
Hemodialysis ^a	Hemofiltration	
<i>N</i> -Acetylcysteine		
Sodium bicarbonate ^b		

^aPotentially deleterious.

^bAs compared with isotonic saline.

N-acetylcysteine (NAC) is an antioxidant with vasodilatory properties that was postulated to potentially prevent CA-AKI based on its capacity to scavenge reactive oxygen species (ROS), reduce the depletion of glutathione, and stimulate the production of vasodilatory mediators, including nitric oxide.^{420,421} Clinical trials of oral and IV NAC have yielded conflicting findings.^{422–431} Although initially used at a dose of 600 mg bid,⁴²² subsequent studies suggested greater efficacy with higher doses of up to 1200 mg bid.^{429,430} In the Acetylcysteine for Contrast-Induced Nephropathy (ACT) trial, 2308 patients were randomized to receive 1200 mg of NAC or placebo bid beginning prior to the procedure and continuing for three doses postprocedure.⁴³¹ No differences were observed in the incidence of CA-AKI at 48 to 96 hours postcontrast administration or in the incidence of death or need for dialysis within 30 days. However, the overall study population had relatively well-preserved kidney function, with a median serum creatinine level of 1.1 mg/dL, and fewer than 16% of patients had a baseline serum creatinine level more than 1.5 mg/dL. In the aforementioned PRESERVE trial, the administration of NAC in a dose of 1200 mg by mouth bid for 5 days beginning just prior to angiography was not associated with a reduction in 90-day death, need for dialysis, or persistent impairment in kidney function (OR, 1.02; 95% CI, 0.78–1.33) or in a reduction in CA-AKI (OR, 1.06; 95% CI, 0.87–1.28).⁴¹⁹ Based on these findings, NAC should not be used to reduce the risk of CA-AKI.

Trials of other pharmacologic interventions, including furosemide, dopamine, fenoldopam, calcium channel blockers, and mannitol, have failed to demonstrate significant benefit and, in some cases, have been associated with an increased risk of CA-AKI.^{432–438} Studies on the benefit of natriuretic peptides, aminophylline, theophylline, and ascorbic acid have also yielded conflicting results.^{439–448} Given the absence of convincing data on the efficacy of these interventions, as well as potential safety concerns with the use of natriuretic peptides, aminophylline, and theophylline in patients with cardiovascular disease, their routine use is not recommended.⁴⁴⁷ There have been multiple trials and meta-analyses that investigated statins for the prevention of CA-AKI. Although many, albeit not all, have demonstrated a benefit of statins, particularly in a high dose, with respect to the development of CA-AKI, the effect of this class of medication on more serious outcomes, including the need for dialysis and progressive CKD, remains unclear.^{449–459} Furthermore, in most patients requiring angiographic procedures, there are likely other indications for statin therapy. RRTs for the prevention of CA-AKI have been largely ineffective and, in some cases, the use of “prophylactic” hemodialysis has been associated with harm.^{460–462} The interpretation of studies of hemofiltration for the prevention of CA-AKI is confounded by their use of change in the serum creatinine level as an endpoint, because hemofiltration lowers serum creatinine concentration.^{463,464} Given the risks associated with IV line placement and the renal replacement procedures themselves, along with lack of definitive benefit, the use of dialysis or hemofiltration to prevent CA-AKI is not currently recommended^{3,465} (Fig. 29.2).

Over the past 25 years, there has been considerable progress in developing less nephrotoxic contrast agents.⁴⁶⁶ The use of lower osmolal contrast agents in place of the older and more nephrotoxic high-osmolal agents has resulted in a

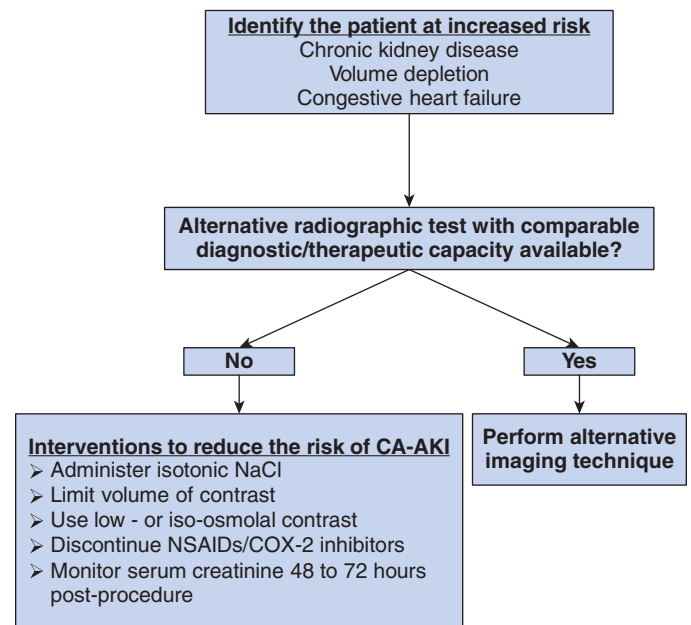


Fig. 29.2 Algorithm for mitigation of risk of contrast-associated acute kidney injury (CA-AKI). COX-2, Cyclooxygenase-2; NSAIDs, nonsteroidal antiinflammatory drugs.

decreased incidence of CA-AKI.^{467,468} Data regarding the added benefit associated with the iso-osmolal radiopaque contrast agent iodixanol has been less consistent^{469–475} and may reflect heterogeneity in the risk of CA-AKI associated with specific lower-osmolal agents.⁴⁷⁶

PREVENTION OF OTHER FORMS OF INTRINSIC ACUTE KIDNEY INJURY

Allopurinol (100 mg/m² every 8 hours, maximum, 800 mg/day) is useful for limiting uric acid generation in patients at high risk for acute urate nephropathy. However, AKI can develop despite the use of allopurinol, probably through the toxic actions of hypoxanthine crystals on tubule function.*

In the setting of high rates of uric acid generation, such as tumor lysis syndrome, the use of recombinant urate oxidase (rasburicase, 0.2 mg/kg) may be more effective. Rasburicase catalyzes the degradation of uric acid to allantoin and has been shown to be effective both as prophylaxis and treatment for acute uric acid-mediated tumor lysis syndrome and to prevent the development of AKI due to tumor lysis syndrome-associated hyperuricemia.^{164,251,478–481} In oligoanuric patients, prophylactic hemodialysis may be used to acutely lower uric acid levels.

Amifostine, an organic thiophosphate, has been demonstrated to ameliorate cisplatin nephrotoxicity in patients with solid organ or hematologic malignancies.^{482–485} NAC limits acetaminophen-induced renal injury if given within 24 hours of ingestion, and dimercaprol, a chelating agent, may prevent heavy metal nephrotoxicity.^{486,487} Ethanol inhibits ethylene glycol metabolism to oxalic acid and other toxic metabolites, but its use has been largely replaced by fomepizole, an

*References 160, 251, 271, 273, 477, and 478.

inhibitor of alcohol dehydrogenase that decreases the production of ethylene glycol metabolites and prevents the development of AKI.^{488–491}

REMOTE ISCHEMIC PRECONDITIONING

Remote ischemic preconditioning (RIPC) has been investigated as a potential intervention for the prevention of AKI. RIPC involves the implementation of brief episode(s) of ischemia and reperfusion of distant tissue—for example, with sequential brief inflation and deflation of a blood pressure tourniquet on a limb that is hypothesized to enhance the kidneys' resistance to a subsequent more prolonged period of ischemia by means of hormonal mediators, hormonal and neuronal signaling pathways, and antiinflammatory and molecular mediators (Fig. 29.3). Recent trials and meta-analyses have examined the benefits of RIPC in the setting of cardiac surgery. A trial that randomized 240 high-risk patients undergoing cardiac surgery to RIPC or sham RIPC has demonstrated a lower rate of AKI, defined by KDIGO criteria, with RIPC (37.5% vs. 52.5%; $P = .02$), with no effect on secondary endpoints, including myocardial infarction, stroke, and death.⁴⁹²

A considerably larger trial that enrolled 1612 patients undergoing cardiac surgery found no difference in the rate of AKI (a secondary endpoint) between RIPC and sham conditioning groups or in the incidence of the combined primary endpoint of death from cardiovascular causes, nonfatal myocardial infarction, coronary revascularization, or stroke within 12 months.⁴⁹³ A series of meta-analyses of studies that included patients undergoing cardiac, and in

some cases, vascular surgery has found lower rates of AKI with RIPC compared with control, but failed to demonstrate a benefit on the need for RRT or death.^{494–496} Based on these cumulative data, the use of RIPC for the prevention of adverse outcomes following cardiac and vascular procedures is not routinely recommended.

PHARMACOLOGIC THERAPY FOR ACUTE TUBULAR NECROSIS

During the past 2 decades, there has been extensive investigation into the pathogenesis of AKI using experimental animal models and cultured cells. These studies have resulted in substantial advances in our understanding of the pathophysiology of ATN in humans and led to the discovery of an array of potentially novel targets for the treatment of this common and serious disease. However, multiple interventions shown to ameliorate AKI in animals have failed to be effective in humans with ATN (Box 29.7). There are many possible reasons for the lack of success in translating therapeutic successes for AKI from animal models to clinical practice. Differences in the cause of ATN in animal models and human disease may contribute to differential responses to pharmacologic therapy. Another principal obstacle relates to the difficulty in identifying the incipient stage of ATN prior to elevations in the serum creatinine concentration or clinical evidence of decreased urine output. Over the past decade, several novel serum and urinary biomarkers have been investigated for their ability to identify AKI in its earliest stages and differentiate ATN from volume-responsive AKI.¹⁷ Work in this

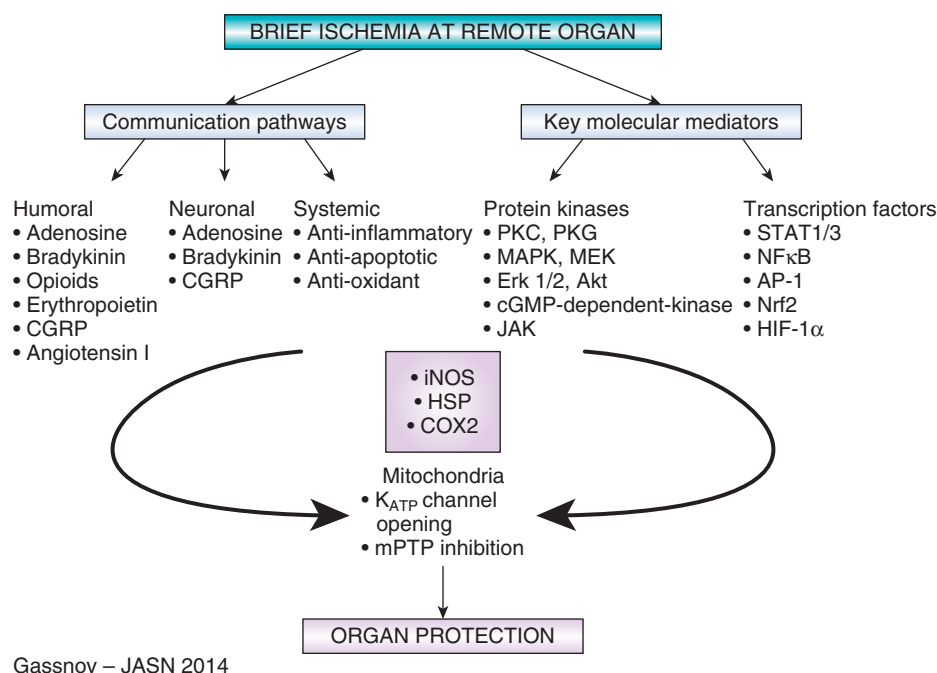


Fig. 29.3 Postulated mechanisms for renal protective effects of remote ischemic preconditioning. *AP-1*, Activator protein-1; *cGMP*, cyclic guanosine monophosphate; *CGRP*, calcitonin gene-related peptide; *COX2*, cyclooxygenase-2; *HIF-1α*, hypoxia-inducible factor 1α; *HSP*, heat shock protein; *iNOS*, inducible nitric oxide synthase; *JAK*, Janus kinase; *MEK*, MAPK kinase; *mPTP*, mitochondrial permeability transition pore; *NFκB*, nuclear factor kappa B; *Nrf2*, nuclear factor (erythroid-derived 2)-like 2; *PKC*, protein kinase C; *PKG*, protein kinase G; *STAT1/3*, signal transducer and activator of transcription. (From Gassnov N, Nia AM, Caglayan E, Er F. Remote ischemic preconditioning and renoprotection: from myth to a novel therapeutic option? *J Am Soc Nephrol*. 2014;25(2):216–224.)

Box 29.7 Pharmacologic Interventions Found to Be Ineffective for Acute Tubular Necrosis in Clinical Trials

Diuretics
Dopamine
Fenoldopam
Thyroid hormone
Alpha melanocyte-stimulating hormone
Atrial natriuretic peptide
Alkaline phosphatase
Insulin growth factor
Erythropoietin
Prostaglandin A1

area may facilitate the identification of those patients most likely to respond to treatments that have been found to be effective in animal models.

DOPAMINE

Historically, low-dose (“renal dose”) dopamine (<2 mg/kg/min) was widely advocated for the management of oliguric AKI.^{497–499} In experimental studies of animals and healthy human volunteers, low-dose dopamine increased renal blood flow and, to a lesser extent, the GFR. However, low-dose dopamine has not been demonstrated to prevent or alter the course of ischemic or nephrotoxic ATN in prospective clinical trials.^{500–504} This absence of clinical benefit may relate to differences in the hemodynamic response to low-dose dopamine in patients with renal disease as compared with healthy individuals. In contrast to the reduction in the renal resistive index associated with low-dose dopamine in critically ill patients without kidney disease, dopamine infusion is associated with an increase in renal resistance in patients with AKI.⁵⁰⁵ Moreover, dopamine, even at low doses, is potentially toxic in critically ill patients and can induce tachyarrhythmias, myocardial ischemia, and extravasation necrosis.⁵⁰⁵ Thus, the routine administration of low-dose dopamine to ameliorate or reverse the course of AKI is not justified based on the balance of experimental and clinical evidence.^{506,507}

FENOLDOPAM

Fenoldopam is a selective postsynaptic dopamine agonist that acts on D1 receptors and mediates more potent renal vasodilation and natriuresis than dopamine.⁵⁰⁸ However, fenoldopam is a potent antihypertensive agent and causes hypotension by decreasing peripheral vascular resistance. Several small studies have suggested that fenoldopam could reduce the incidence of AKI in high-risk clinical situations^{509,510}; however, a subsequent larger randomized trial comparing fenoldopam with standard hydration in patients undergoing invasive angiographic procedures found no benefit in regard to decreasing the incidence of CA-AKI.⁴³⁵ In another large RCT, fenoldopam administration failed to reduce mortality or the need for renal replacement therapy in ICU patients with early ATN.⁵¹¹ Therefore, there is currently no clinical role for fenoldopam in the prevention or treatment of AKI.

NATRIURETIC PEPTIDES

ANP is a 28-amino acid polypeptide synthesized in cardiac atrial muscle.^{512,513} ANP augments the GFR by triggering afferent arteriolar vasodilation and constriction of the efferent arteriole.^{514,515} In addition, ANP inhibits sodium transport and lowers oxygen requirements in several nephron segments.^{516,517} Synthetic analogues of ANP have shown promise in the management of ATN in the laboratory setting; however, these benefits in animal models of AKI have failed to translate into clinical benefit in humans. A large multicenter, prospective, randomized, placebo-controlled trial of anaritide, a synthetic analogue of ANP, in patients with ATN failed to show clinically significant improvement in dialysis-free survival or overall mortality,⁵¹⁸ although there was an improvement in dialysis-free survival in oliguric patients. This benefit in oliguric patients was not confirmed in a subsequent prospective study.⁵¹⁹ It has been suggested that the absence of benefit may be related to both the relatively late initiation of therapy and to the effect of ANP on systemic blood pressure. In a subsequent pilot study, low-dose recombinant ANP administration in high-risk cardiac surgery patients was associated with a reduction in the requirement for postoperative RRT.⁵²⁰ Until these results are confirmed in a larger, multicenter trial, the use of ANP in this setting cannot be recommended. Trials of ANP for the prevention of contrast-associated AKI have generated mixed results.^{444,445} Ularitide (urodilantin) is a natriuretic pro-ANP fragment produced in the kidney. In a small randomized trial, ularitide did not reduce the need for dialysis in patients with AKI.⁵²¹ A meta-analysis of ANP for the treatment of AKI has concluded that the paucity of high-quality studies precludes a determination of the effects of this therapy.⁵²²

LOOP DIURETICS

High-dose IV diuretics are commonly prescribed to increase urine output in patients with oliguric AKI. Although this strategy assists in volume management and minimizes the risk of progressive volume overload, there is no evidence that diuretic therapy alters the natural history of AKI or improves mortality or dialysis-free survival. In a retrospective analysis, diuretic therapy was associated with an increased risk of death and nonrecovery of renal function.⁵²³ These risks were restricted, however, to patients who did not respond to diuretic administration with increased urine volume; in diuretic-responsive patients, outcomes were similar to those of untreated patients. In a prospective randomized trial, high-dose IV furosemide augmented urine output but did not alter the outcome of established AKI.⁵²⁴ In a post hoc analysis of data from the Fluid and Catheter Treatment Trial, a positive fluid balance after AKI in patients with acute lung injury was strongly associated with increased mortality, whereas diuretic therapy was associated with improved 60-day patient survival.⁵²⁵ Given the risks of loop diuretics in AKI, including irreversible ototoxicity and exacerbation of prerenal AKI, these agents should be used solely to facilitate the management of extracellular volume overload (see later).⁵²⁶ Of note, a single administration of furosemide in a dose of 1.0 to 1.5 mg/kg has been shown to help characterize the risk for progressive AKI.⁵²⁷ Specifically, a urine volume less than 200 mL in the 2 hours following furosemide administration demonstrated sensitivity and specificity for progression to AKIN stage 3 of 87.1% and 84.1%, respectively.

MANNITOL

The osmotic diuretic mannitol, which also has renal vasodilatory and oxygen-free radical scavenging properties, has been investigated as a preventive treatment for AKI.^{528,529} No adequate data exist to support the routine administration of mannitol to oliguric patients. Moreover, when administered to severely oliguric or anuric patients, mannitol may trigger an expansion of intravascular volume and pulmonary edema, as well as severe hyponatremia due to an osmotic shift of water from the intracellular to the intravascular space.⁵³⁰

MANAGEMENT OF OTHER CAUSES OF INTRINSIC ACUTE KIDNEY INJURY

ACUTE VASCULITIS AND ACUTE GLOMERULAR DISEASE

The management of acute vasculitis involving the kidney and acute glomerular disease is covered in detail in Chapters 31 and 32. AKI caused by acute glomerulonephritis or vasculitis may respond to corticosteroids, alkylating agents, rituximab, and plasmapheresis, depending on the primary cause of the disease. Plasma exchange is useful in the treatment of sporadic TTP and possibly sporadic HUS in adults.^{531,532} The role of plasmapheresis in drug-induced thrombotic microangiopathies is less certain, and removal of the offending agent is the most important initial therapeutic maneuver.^{144,533,534} Postdiarrheal HUS in children is usually managed conservatively because studies have shown that early antibiotic therapy may actually promote the development of HUS.⁵³⁵ Treatment with eculizumab, a humanized monoclonal antibody that prevents cleavage of complement component C5 into C5a and C5b, inhibiting terminal complement activation, may be considered in patients with nondiarrheal (complement-mediated) HUS unresponsive to plasma exchange.⁵³⁶ Hypertension and AKI associated with scleroderma may be exquisitely sensitive to treatment with ACE inhibitors.^{537–539}

ACUTE KIDNEY INJURY IN MULTIPLE MYELOMA

Early studies have suggested that plasmapheresis may be of benefit in AKI due to myeloma cast nephropathy.^{169,540,541} Clearance of circulating light chains, with concomitant chemotherapy to decrease the rate of production, had been postulated to reverse renal injury in patients with circulating light chains, heavy Bence Jones proteinuria, and AKI. A subsequent RCT compared plasma exchange and standard chemotherapy with chemotherapy alone. Although the study did not demonstrate improvement with plasma exchange with regard to a composite outcome of death, dialysis dependence, or a GFR less than 30 mL/min/1.73 m² at 6 months, the study was inadequately powered to exclude a clinical benefit definitively, and there was a trend toward improved outcomes with plasmapheresis.^{170–172} More recently, it has been suggested that the use of dialysis membranes that are permeable to light chains and other proteins with molecular weights lower than albumin (high-cutoff membranes) may be an effective therapeutic strategy in patients with AKI due to light chain cast nephropathy; however, data from clinical trials have yielded inconclusive results.^{173–175} Thus, primary management should focus on the prompt initiation of highly effective chemotherapy. Other contributors to AKI in multiple

myeloma, such as hypercalcemia, should also be promptly treated.

ACUTE INTERSTITIAL NEPHRITIS

Most cases of acute interstitial nephritis (AIN) are due to an allergic response to a medication.⁵⁴² The initial therapeutic step in AIN is discontinuation of the offending medication or treatment of the probable inciting factor if not drug-induced. Data on the efficacy of corticosteroids have been derived from small observational studies, which have yielded highly discordant results. Although some studies have suggested that early use of corticosteroids (i.e., prior to significant renal damage and within 7–14 days of discontinuation of the offending medication)⁵⁴³ may be beneficial, other studies have demonstrated no clear evidence of efficacy.⁵⁷ There have been no large, prospective RCTs investigating the role of corticosteroids in the treatment of AIN. Because corticosteroids are associated with a series of potentially serious side effects, their use should be considered on a case by case basis. If corticosteroid therapy is being considered and no patient-related contraindications exist, one potential regimen used in one study involved the IV administration of methylprednisolone (250–500 mg/day) for 3 to 4 days followed by oral prednisone at a dose of 1 mg/kg/day tapered over 8 to 12 weeks.⁵⁴³ However, there are no data supporting the superiority of this specific approach over others. Mycophenolate mofetil has also been investigated as a therapeutic agent for AIN. In a study of eight patients with AIN, six experienced improvement and two experienced stabilization in kidney function with mycophenolate mofetil therapy.⁵⁴⁴ Although this small case series suggests a possible role for mycophenolate mofetil in the treatment of AIN, additional studies are needed to confirm its safety and efficacy for this indication.

NONDIALYTIC SUPPORTIVE MANAGEMENT OF ACUTE KIDNEY INJURY–ASSOCIATED COMPLICATIONS

Metabolic complications such as intravascular volume overload, hyperkalemia, hyperphosphatemia, and metabolic acidosis are common in oliguric AKI, and preventive measures should be implemented, beginning with the initial diagnosis (see Table 29.7). Adequate nutrition should be provided to meet caloric requirements and minimize catabolism. In addition, all medications that are normally excreted by the kidney need to be adjusted based on the severity of the renal impairment.

EXTRACELLULAR VOLUME OVERLOAD

After correction of intravascular volume deficits, salt and water intake should be adjusted to match ongoing losses—urinary, gastrointestinal, drainage sites, insensible losses. Extracellular volume overload can usually be managed by restriction of salt and water intake and by judicious use of diuretics. High doses of loop diuretics (e.g., the equivalent of 200 mg of furosemide administered as an IV bolus infusion or 20 mg/hour as a continuous infusion) or combination therapy with both thiazide and loop diuretics may be required. If an adequate diuresis cannot be attained, further use of diuretics should be discontinued to minimize the risk of complications, such as ototoxicity. Fluid administration should be closely monitored to avoid progressive volume overload. Although there is a

strong association between progressive fluid overload and mortality risk in patients with AKI,^{545–547} a causal relationship has not been definitively established because volume overload may also be a surrogate for other determinants of mortality, such as hemodynamic instability and capillary leak. Fluid conservative management has, however, been demonstrated to result in improved outcomes in critically ill patients with respiratory failure.⁵⁴⁸ Ultrafiltration or dialysis may be required for volume management when conservative measures fail.

HYPONATREMIA AND HYPERNATREMIA

Hyponatremia associated with a fall in effective serum osmolality can usually be corrected by restriction of water intake. Conversely, hypernatremia is treated by the administration of water, hypotonic saline solutions, or hypotonic dextrose-containing solutions (the latter are effectively hypotonic because dextrose is rapidly metabolized).

HYPERKALEMIA

Mild hyperkalemia (<5.5 mmol/L) should be managed initially by restriction of dietary potassium intake and the discontinuation of potassium supplements and potassium-sparing diuretics. More severe degrees of hyperkalemia (5.5–6.5 mmol/L) can usually be controlled by combining these measures with the administration of exchange resins to enhance gastrointestinal potassium losses. Although sodium polystyrene sulfonate has been widely used for decades, concerns have been raised regarding its safety, particularly when administered in 70% sorbitol, due to reports of bowel necrosis.^{549,550} Newer exchange resins, patiromer and zirconium cyclosilicate, are also effective at decreasing the serum potassium concentration, although patiromer is not labeled for the acute management of hyperkalemia. Loop diuretics can also increase potassium excretion in diuretic-responsive patients. Emergency measures need to be used in patients with more severe hyperkalemia and in patients with electrocardiographic manifestations of hyperkalemia. In patients with severe hyperkalemia with concomitant electrocardiographic manifestations, the IV administration of calcium will antagonize the cardiac and neuromuscular effects of hyperkalemia and is a valuable emergency temporizing measure, allowing time for the additional measures described later to be implemented. IV calcium must be used with caution, however, if there is concomitant severe hyperphosphatemia or evidence of digitalis toxicity. IV insulin (10–20 U of regular insulin) promotes potassium entry into cells and lowers extracellular potassium concentration within 15 to 30 minutes, with an effect that lasts for several hours.^{551,552} The concomitant administration of IV dextrose (25–50 g over 30–60 minutes) is required to prevent hypoglycemia in patients who do not have hyperglycemia. Beta-adrenergic agonists, such as inhaled albuterol (10–20 mg by nebulizer), also promote rapid potassium uptake into the intracellular compartment.⁵⁵¹ Although sodium bicarbonate also stimulates potassium uptake into the intracellular compartment, this effect is not sufficiently rapid to be clinically useful for the emergent management of hyperkalemia.⁵⁵² Emergent dialysis is indicated if hyperkalemia is resistant to these measures.

METABOLIC ACIDOSIS

The treatment of metabolic acidosis is dependent on the clinical setting and cause. As a general rule, metabolic acidosis

does not require emergent treatment unless the serum HCO_3^- concentration falls below 15 mmol/L or the pH is lower than 7.15 to 7.20. In patients with AKI in whom metabolic acidosis is due to the underlying renal failure, more severe acidosis can be corrected by oral or IV bicarbonate administration. Initial rates of replacement should be based on estimates of the HCO_3^- deficit and adjusted thereafter according to serum levels. In patients with underlying lactic acidosis, the role of bicarbonate therapy is controversial, and the primary focus of therapy should be on correction of the underlying cause.^{553–556} Patients treated with IV bicarbonate need to be monitored for complications of therapy, including metabolic alkalosis, hypocalcemia, hypokalemia, hypernatremia, and volume overload.

DISTURBANCES OF CALCIUM, PHOSPHATE, MAGNESIUM, AND URIC ACID

Hypocalcemia does not usually require treatment unless it is severe or symptomatic, as may occur in patients with rhabdomyolysis or pancreatitis or after the administration of bicarbonate. Hyperphosphatemia can often be controlled by restricting dietary phosphate intake and the use of oral phosphate binders (e.g., aluminum hydroxide, calcium salts, sevelamer carbonate, lanthanum carbonate). Caution should be used with aluminum-containing phosphate binders because prolonged use may result in aluminum intoxication, which can contribute to osteomalacia; short-term use is rarely associated with bone disease, and the feared neurologic complications of aluminum intoxication are restricted to patients with inadvertent parenteral exposure. Hypermagnesemia can be prevented through avoidance of magnesium-containing medications, such as antacids, and limiting the magnesium content of parenteral nutrition. Hyperuricemia is usually mild in AKI (<15 mg/dL) and does not require a specific intervention. Severe hyperuricemia secondary to cell lysis may be managed by blocking xanthine oxidase with allopurinol or by enhancing degradation with recombinant uricase, as previously described.

NUTRITIONAL MANAGEMENT

Patients with AKI are clinically heterogeneous, and individualized nutritional management is required, especially in critically ill patients on RRT in whom protein catabolic rates can exceed 1.5 g/kg body weight/day.^{3,285,286,288,289,557,558}

The objective of nutritional management in AKI is to provide sufficient calories to preserve lean body mass, avoid starvation ketoacidosis, and promote healing and tissue repair while minimizing production of nitrogenous waste. If the duration of impaired kidney function is likely to be short, the patient is not extremely catabolic and does not require RRT, then dietary protein should be approximately 0.8 to 1.0 g/kg body weight/day.³ Protein intake should not be restricted in patients in whom AKI is likely to be prolonged, are hypercatabolic, or are receiving RRT. Protein intake in these patients should generally be 1.0 to 1.5 g/kg body weight per day.^{285,286,557,558} There has been no evidence of improved outcomes with protein intake higher than 1.7 g/kg body weight per day, even in extremely hypercatabolic patients.³ Total caloric intake should generally be 20 to 30 kcal/kg body weight per day and should not exceed 35 kcal/kg per day.^{3,285,286,557,558} Benefits of vigorous parenteral hyperalimentation have not been consistently demonstrated;

enteral nutrition support is preferred because it avoids the morbidity associated with parenteral nutrition while providing support to intestinal function.²⁸⁸ Water-soluble vitamins and trace elements should be supplemented in patients receiving RRT.^{557,558}

ANEMIA

Severe anemia is generally managed with a blood transfusion. Transfusion is usually not required for patients with a hemoglobin concentration above 7 g/dL.⁵⁵⁹ Whether there is a role for erythropoiesis-stimulating agents in AKI has not been definitively determined.⁵⁶⁰ Patients with AKI or another acute illness are relatively resistant to the effects of these agents, and their onset of action is delayed. In RCTs in critically ill patients, recombinant human erythropoietin decreased transfusion requirement but had no effect on other outcomes.^{561,562} Uremic bleeding usually responds to desmopressin, correction of anemia, estrogens, or dialysis.

DRUG DOSING

Doses of drugs that are excreted by the kidney must be adjusted for impaired kidney function and the use of RRT.^{563–565} Whenever possible, pharmacokinetic monitoring should be used to ensure appropriate drug dosing, especially for agents with narrow therapeutic windows (see Chapter 61). In addition to careful monitoring for toxicity of agents that are normally excreted by the kidney, careful attention must be paid to dosing of antibiotics and other drugs removed by RRT to ensure that therapeutic drug levels are achieved, particularly in patients receiving augmented intensity of RRT.

RENAL REPLACEMENT THERAPY IN ACUTE KIDNEY INJURY

GENERAL PRINCIPLES

RRT is the generic term for the multiple modalities of dialysis and hemofiltration used in the management of kidney failure. Although kidney transplantation is also a form of RRT for ESKD, transplantation does not play a role in the management of AKI. Renal replacement therapy facilitates the management of patients with AKI, allowing correction of acid-base and electrolyte disturbances, amelioration of volume overload, and removal of byproducts of nitrogen metabolism (so-called uremic solutes). Although RRT can forestall or reverse the life-threatening complications of uremia associated with severe and prolonged AKI, it does not hasten and can potentially delay the recovery of kidney function in patients with AKI,⁵⁶⁶ and can itself be associated with potentially life-threatening complications.⁵⁶⁷ Despite more than 60 years of research and clinical experience,^{568,569} numerous questions regarding the optimal application of RRT in AKI remain.^{3,570–573}

INDICATIONS FOR AND TIMING OF INITIATION OF RENAL REPLACEMENT THERAPY

In clinical practice, there are wide variations in the timing of initiation of RRT for patients with AKI.⁵⁷⁴ Widely accepted indications for initiation of RRT include volume overload unresponsive to diuretic therapy, severe metabolic acidosis or hyperkalemia, despite appropriate medical therapy, and overt uremic manifestations, including encephalopathy, pericarditis, and uremic bleeding diathesis (Box 29.8). However, even these specific indications are subject to

Box 29.8 Indications for Renal Replacement Therapy

Absolute Indications

- Volume overload unresponsive to diuretic therapy
- Persistent hyperkalemia despite medical therapy
- Severe metabolic acidosis
- Overt uremic symptoms—encephalopathy, pericarditis, uremic bleeding diathesis

Relative Indications

- Progressive azotemia without uremic manifestations
- Persistent oliguria

substantial clinical interpretation. In many patients, RRT is initiated in the absence of these specific indications in response to a clinical course marked by progressive azotemia or sustained oliguria. The correlation between the BUN concentration and the onset of uremic symptoms is relatively weak; although the longer the duration and the greater the severity of azotemia, the more likely it is that overt symptoms will develop. Observational series and small clinical trials dating from the 1950s through the 1980s have suggested that initiating RRT when the BUN concentration approached 90 to 100 mg/dL was associated with improved survival as compared with more delayed initiation of therapy.^{575–579} Other observational studies have suggested that the initiation of RRT at an even less severe degree of azotemia may further improve survival.^{580–583} These studies need to be interpreted with caution, however, because the outcomes associated with earlier initiation of RRT may reflect differences related to the reasons for initiation of therapy (e.g., volume overload or hyperkalemia vs. progressive azotemia) rather than a benefit due to the earlier therapy per se. In addition, these observational series only included patients in whom RRT was actually initiated, rather than the broader population of patients with AKI, including patients who recovered kidney function or died without receiving RRT.

There have been an increasing number of prospective clinical trials evaluating the timing of initiation of RRT in AKI. In a small RCT of critically ill patients randomized to early, high-volume hemofiltration, early low-volume hemofiltration, or late low-volume hemofiltration, there was no benefit associated with earlier initiation of treatment.⁵⁸⁴ In a subsequent trial comparing earlier to later initiation of dialysis in patients with community-acquired AKI, mortality was lower in patients initiated on dialysis later than in the group of patients started earlier, with no difference in recovery of kidney function between groups.⁵⁸⁵ This latter trial needs to be interpreted with caution, however, because almost half of the patients admitted with community-acquired AKI were excluded due to the urgent need for dialysis.

More recently, two clinical trials have compared differing strategies of early and delayed initiation of RRT in patients with AKI. The Effect of Early vs. Delayed Initiation of Renal Replacement Therapy on Mortality in Critically Ill Patients with Acute Kidney Injury (ELAIN) trial randomized 231 critically ill patients with AKI (based on KDIGO stage 2 and plasma neutrophil gelatinase-associated lipocalin level >150 ng/mL)

at a single center to early or delayed initiation of RRT; it demonstrated reduced 90-day mortality (hazard ratio, 0.66; 95% CI, 0.45–0.97) with earlier initiation of RRT.⁵⁸⁶ Patients in the early RRT group also demonstrated decreased duration of RRT (9 vs. 25 days; $P = .04$) and reduced hospital length of stay (51 vs. 82 days; $P < .001$). However, the separation in timing of the initiation of RRT between early and late groups was less than 1 day, raising questions about the therapeutic mechanisms underlying the observed marked reductions in mortality, duration of RRT, and hospital length of stay. It is possible that unrecognized differences between the two treatment groups may have contributed to the surprisingly large effect size.

The contemporaneous Artificial Kidney Initiation in Kidney Injury (AKIKI) trial was a multicenter trial that randomized 620 patients with KDIGO stage 3 AKI who required mechanical ventilation and/or catecholamine support to early or delayed initiation of RRT. It found no difference in 60-day mortality between the groups (48.5% in the early strategy group vs. 49.7% in the delayed strategy group; $P = .79$)⁵⁸⁷ (Fig. 29.4). Notably, nearly half ($n = 151$) of patients assigned to the delayed strategy group never required initiation of RRT. Subsequently, the Initiation of Dialysis Early Versus Delayed in the Intensive Care Unit (IDEAL-ICU) trial also failed to demonstrate a benefit to the earlier initiation of RRT in 488 patients with sepsis-associated AKI.⁵⁸⁸ In the IDEAL-ICU trial, patients were enrolled if they met the criteria for RIFLE-F AKI and had sepsis and did not have an emergent indication for RRT. In the early-treatment arm, patients initiated RRT within 12 hours of eligibility while in the delayed-treatment arm; RRT was initiated if a specific indication for RRT developed or if there was no recovery of kidney function after 48 hours. In the early treatment arm, mortality at 90 days was 58% as compared with 54% in the delayed treatment arm ($P = .38$). Of patients in the early treatment arm, 97% received RRT as compared with only 62% of patients in the delayed treatment arm. Of those in the delayed arm who did not receive RRT, 75% had spontaneous recovery of kidney

function, whereas 23% died prior to specified criteria for the initiation of RRT.

Key differences between the ELAIN trial and AKIKI and IDEAL-ICU trials should be noted. In particular, the entry criterion for both the AKIKI and IDEAL-ICU trials (stage 3 or RIFLE-F AKI) was the trigger for delayed initiation of RRT in the ELAIN trial. In addition, both the AKIKI and IDEAL-ICU trials excluded individuals with an emergent indication for RRT, in whom a strategy of delayed therapy would be inappropriate. The ongoing Standard vs Accelerated Initiation of RRT in AKI (START-AKI) trial (NCT02568722) may help reconcile the divergent results between the ELAIN trial and AKIKI and IDEAL-ICU trials and clarify the benefit of early initiation of RRT in patients with AKI who do not have overt clinical or biochemical indications for this treatment.⁵⁸⁹

Although volume overload unresponsive to diuretic therapy is a widely accepted indication for the initiation of RRT, there are wide variations in the degree of volume overload at initiation of therapy.^{546,590,591} Observational studies have demonstrated a strong association between the degree of volume overload and mortality risk, leading to the suggestion that RRT should be initiated early, prior to the development of progressive volume overload.^{545,592} It should be recognized, however, that the association between volume overload and mortality risk does not establish a causal relationship; disease processes that contribute to the development of volume overload may independently contribute to mortality risk in these patients. Prospective studies will therefore be required to demonstrate that preemptive RRT, prior to the development of more severe degrees of volume overload, decreases morbidity and mortality.

Given the current level of evidence, the KDOQI clinical practice guideline for acute kidney injury does not make strong recommendations on the timing of initiation of RRT.³ It suggests that RRT be “...initiated emergently when life-threatening changes in fluid, electrolyte, and acid-base balance exist”,³ and that “...the broader clinical context, the presence

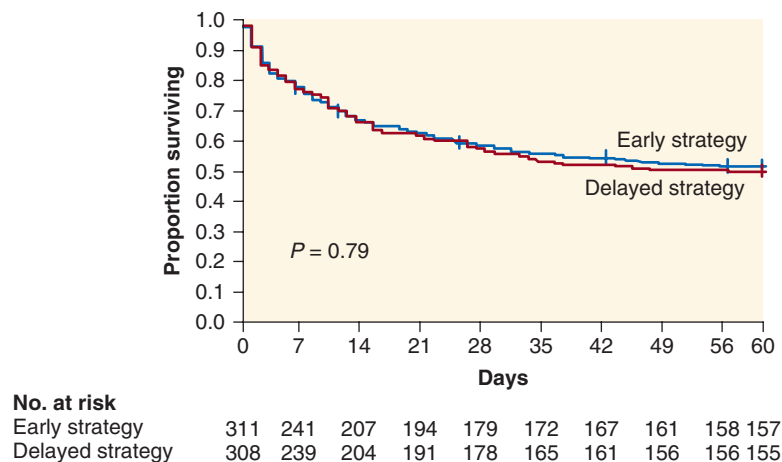


Fig. 29.4 Probability of survival and timing of treatment with early versus delayed strategies for initiation of renal replacement therapy (RRT). Kaplan-Meier probability of survival and timing of initiation of RRT in the Acute Kidney Initiation in Kidney Injury (AKIKI) trial. In the early-treatment group, 60-day mortality was 48.5% versus 49.7% in the delayed-treatment group (hazard ratio, 1.03; 95% confidence interval, 0.81–1.29; $P = .84$). In the early-treatment group, 98% of patients initiated RRT at a median of 4.3 hours after reaching stage 3 AKI as compared with 51% of patients who initiated RRT at a median of 57 hours. (From Gaudry S, Hajage D, Schortgen F, et al. Initiation strategies for renal-replacement therapy in the intensive care unit. *N Engl J Med*. 2016;375:122–133.)

of conditions that can be modified by RRT, and trends of laboratory tests—rather than single BUN and creatinine thresholds alone [be considered] when making the decision to start RRT.”³

RRT should be discontinued when kidney function recovers or because the continued provision of dialytic support is no longer consistent with the patient’s overall goals of care.³ Recovery of kidney function is usually heralded by increased urine volume. Although no specific threshold of urine output correlates with sufficient recovery of kidney function, it is unlikely that a urine output of less than roughly 1 L/day is sufficient to sustain dialysis independence. Although diuretics may increase daily urine volume, there is no evidence that diuretic therapy promotes recovery of kidney function.⁵⁹³ Improved solute clearance is manifested by a spontaneous fall in blood urea and creatinine concentrations or a persistent downward trend in predialysis values. The role of creatinine clearance measurement to assess the recovery of kidney function is uncertain, with a paucity of data to define specific thresholds for recovery of kidney function. In the Acute Renal Failure Trial Network Study, RRT was continued if measured creatinine clearance on a 6-hour timed urine collection was less than 12 mL/min, RRT was stopped if the clearance was more than 20 mL/min, and the decision was left to the discretion of the clinician if the creatinine clearance was between 12 and 20 mL/min.^{594,595}

CHOICE OF MODALITY OF RENAL REPLACEMENT THERAPY

Multiple modalities of RRT are available for the management of patients with AKI, including conventional intermittent hemodialysis (IHD), peritoneal dialysis (PD), multiple forms of continuous renal replacement therapy (CRRT) and prolonged intermittent renal replacement therapies (PIRRT) such as sustained low-efficiency dialysis (SLED; also known as extended-duration dialysis, EDD). Detailed descriptions of the technical aspects of these modalities are provided in Chapters 63, 64, and 65. Objective data to guide the selection of modality for individual patients are limited, and the choice of modality is often guided by the resources of the health care institution and the technical expertise of the physicians and nursing staff. The KDIGO Clinical Practice Guideline for Acute Kidney Injury suggests that for most patients, the available modalities of RRT are complementary, with the caveats that CRRT and PIRRT be used in hemodynamically unstable patients and that CRRT be used for patients with acute brain injury or other causes of increased intracranial pressure or generalized brain edema.³

Intermittent Hemodialysis

IHD has been the mainstay of RRT in AKI for more than 6 decades. Patients typically undergo dialysis treatments for 3 to 5 hours on a thrice-weekly, alternate-day, or daily schedule, depending on catabolic demands, electrolyte disturbances, and volume status. Just as with the timing of initiation of dialysis in AKI, the most appropriate dosing strategy for IHD in patients with AKI has been the subject of considerable investigation. The dose of IHD may be adjusted by altering the intensity of each individual dialysis session, usually quantified as the product of urea clearance and dialysis duration normalized to the volume of distribution of urea (Kt/V_{urea}) or by changing the frequency of the dialysis sessions. In an

observational study, Paganini and colleagues have demonstrated a survival benefit for patients with intermediate severity of illness scores when the delivered Kt/V_{urea} was more than 1.0 per treatment as compared with a delivered Kt/V_{urea} less than 1.0 per treatment.⁵⁹⁶

However, there have been no prospective clinical trials evaluating the relationship between the delivered Kt/V_{urea} and outcomes when dialysis is provided on a constant treatment schedule. Schiffl and colleagues have reported on a prospective trial of 160 patients with AKI assigned in an alternating fashion to alternate-day or daily intermittent hemodialysis.⁵⁹⁷ The more frequent treatment schedule was associated with a reduction in mortality at 14 days after the last dialysis session, from 46% in the alternate-day dialysis arm to 28% in the daily treatment arm ($P = .01$). The duration of renal failure declined from 16 ± 6 to 9 ± 2 days ($P = .001$). This study has been criticized, however, because the delivered dose of therapy per session was low in both treatment arms ($[Kt/V_{urea}] < 0.95$), resulting in a high rate of symptoms in the alternate-day dialysis arm that may have been related to overtly inadequate dialysis.⁵⁹⁸

The impact of frequency of IHD was also evaluated in the Acute Renal Failure Trial Network study.⁵⁹⁴ In this study, 1124 critically ill patients were randomized to an intensive or less intensive strategy for the management of RRT. When patients were hemodynamically stable, they received IHD and, when hemodynamically unstable, they received CRRT or sustained low-efficiency dialysis (SLED), regardless of treatment arm. Patients randomized to the less intensive treatment strategy received IHD on a thrice-weekly (alternate-day except Sunday) schedule while patients randomized to the intensive arm received IHD six times per week (daily except Sunday). All-cause mortality at 60 days was 53.6% in the intensive treatment arm compared with 51.5% in the less intensive arm ($P = .47$)⁵⁹⁴ (Fig. 29.5). The mean delivered Kt/V_{urea} was 1.3/treatment after the first IHD session. Although the study was not designed to evaluate outcomes by individual modality of RRT, there were no differences in mortality between groups when evaluated based on the percentage of time treated using IHD.⁵⁹⁹

Based on these results, it does not appear that there is further benefit to increasing the frequency of IHD treatments routinely beyond three times per week as long as the delivered Kt/V_{urea} is at least 1.2 per treatment. More frequent treatments may be necessary if the target dose per treatment cannot be achieved—for example, in hypercatabolic patients, in patients with severe hyperkalemia or metabolic acidosis, and for issues related to volume management. The KDIGO Clinical Practice Guideline for Acute Kidney Injury recommends delivering a Kt/V_{urea} of 3.9 per week when using IHD in AKI, calculating the weekly Kt/V_{urea} as the arithmetic sum of the delivered dose per treatment.³ It should be recognized, however, that this approach for calculating an equivalent weekly Kt/V_{urea} is not consistent with urea kinetic principles, and that rigorous data for the appropriate dose of therapy when treatments are delivered more frequently than three times per week are not available.¹³

The selection of IHD dialyzer membrane may also affect clinical outcomes. Exposure to cellulosic membranes results in accentuated leukocyte and complement activation and delayed recovery of kidney function in experimental models of AKI as compared with exposure to more biocompatible

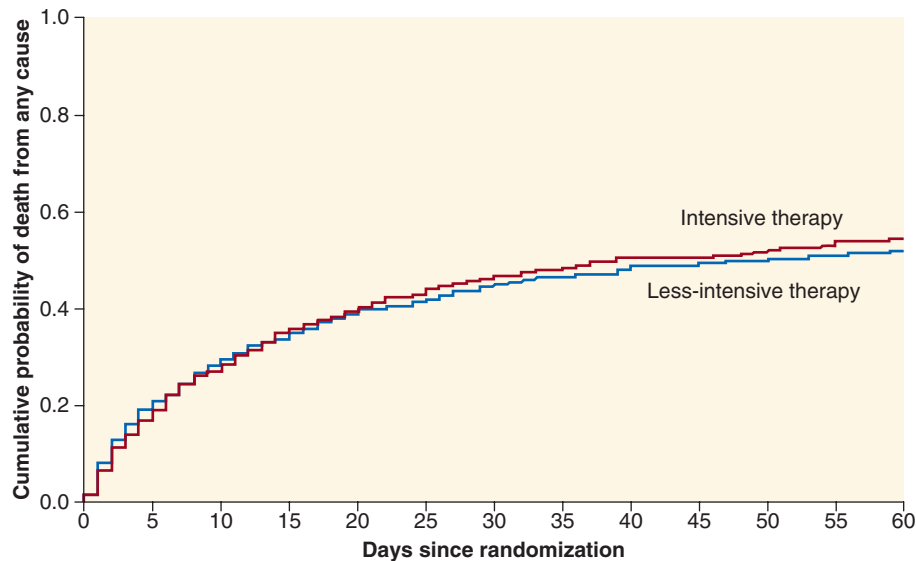


Fig. 29.5 60-Day mortality with intensive versus less intensive renal replacement therapy in the Acute Renal Failure Trial Network (ATN) study. Shown is the Kaplan-Meier plot of mortality in 1124 critically ill patients with acute kidney injury randomized to a strategy of more intensive renal replacement therapy (RRT; 6 times per week intermittent hemodialysis or continuous venovenous hemodiafiltration at 35 mL/kg per hour) versus less intensive RRT (3 times per week intermittent hemodialysis or continuous venovenous hemodiafiltration at 20 mL/kg per hour). At 60 days, mortality was 53.6% in the more intensive arm versus 51.5% in the less intensive arm (odds ratio, 1.09; 95% confidence interval, 0.86–1.40; $P = .47$). (From VA/NIH Acute Renal Failure Trial Network; Palevsky PM, Zhang JH, O'Connor TZ, et al. Intensity of renal support in critically ill patients with acute kidney injury. *N Engl J Med*. 2008;359:7–20.)

synthetic membranes.^{600,601} Clinical trials comparing dialysis membranes have yielded conflicting results. Although some studies have demonstrated delayed recovery of kidney function with cellulosic membranes,^{602–604} other studies have observed no difference between cellulosic and other synthetic membranes thought to be more biocompatible.^{605–609} When these data have been aggregated in systematic reviews, a benefit of the synthetic membranes is not convincingly demonstrated.^{610,611} Although the effect of membrane type on humoral and cellular activation may still influence recovery of kidney function in AKI, the clinical importance of this issue has diminished as the cost differential between synthetic and cellulosic membranes has narrowed, and the use of unsubstituted cellulosic membranes has decreased.

The major complications associated with IHD are related to the need to access the vasculature, the need for anticoagulation to maintain patency of the extracorporeal circuit, and intradialytic hypotension primarily resulting from shifts in solute and volume.^{594,597,612} Many of these issues, particularly the need for vascular access and anticoagulation, are similar for CRRT and SLED.

Vascular access is usually obtained through the insertion of a double-lumen catheter into a large caliber central (internal jugular or subclavian) or femoral vein.⁶¹³ The major complications associated with vascular access include vascular and organ trauma during insertion, bleeding, catheter malfunction and thrombosis, and infection.⁶¹³ Although femoral catheters are generally associated with an increased risk of infection compared with catheters in the subclavian or internal jugular veins, an increased risk of infection was observed only when femoral vein catheters were used in patients with a high body mass index (BMI) in an RCT involving patients undergoing acute RRT.⁶¹⁴ The more prompt transition to tunneled hemodialysis

catheters (or placement of tunneled hemodialysis catheters in advance of initiation) has been proposed as a means of decreasing the risk of infection in patients undergoing acute dialysis.^{615,616} However, this strategy has not been rigorously evaluated in prospective clinical trials.

Anticoagulation is used to help maintain patency of the extracorporeal dialysis circuit in IHD, as well as in CRRT and SLED.^{617,618} The most commonly used anticoagulant for dialysis is unfractionated heparin; with multiple protocols used to attain sufficient anticoagulation of the dialysis circuit while minimizing systemic effects.^{617,618} Regional heparinization, in which heparin is infused proximally to the dialyzer, and protamine is infused into the return line to reverse heparin's effect,⁶¹⁹ can be used but has generally been supplanted by low-dose heparin protocols.⁶²⁰ Low-molecular-weight heparin (LMWH) may be used as an alternative to unfractionated heparin; however, the benefits of this approach are unclear because LMWH is not associated with enhanced efficacy, drug half-life is variably prolonged with impaired kidney function, and monitoring of the anticoagulant effect is more difficult.⁶¹⁷ In patients with heparin-induced thrombocytopenia (HIT), heparin administration is contraindicated. Alternative anticoagulation strategies include regional citrate,^{617,621–623} the serine protease inhibitor nafamostat,⁶²⁴ the direct thrombin inhibitors hirudin, lepirudin, and argatroban^{625–629} and, rarely, the prostanoids epoprostenol and iloprost.^{617,618} In many patients, particularly those with underlying coagulopathy or thrombocytopenia, and in patients with active hemorrhage or recent postoperative status, acute RRT can be provided in the absence of anticoagulation.^{594,630,631}

Intradialytic hypotension is common in patients undergoing acute IHD.^{566,594,599,612,632} Episodes of hypotension may impair solute clearance and the efficiency of dialysis and can further

compromise renal perfusion and delay recovery of kidney function.^{566,633–635} Intradialytic hypotension is typically triggered by intercompartmental fluid shifts or excessive fluid removal, leading to decreased intravascular volume, and may be exacerbated by altered vascular responsiveness related to the underlying acute process.^{612,636} Hypotension may be particularly problematic in critically ill patients in whom sepsis, cardiac dysfunction, hypoalbuminemia, malnutrition, or large third space losses may accompany the development of AKI. The prevention of intradialytic hypotension requires careful assessment of intravascular volume, prescription of realistic ultrafiltration targets, extension of treatment time so as to minimize the ultrafiltration rate, increasing the dialysate sodium concentration, and decreasing the dialysate temperature.^{632,636–638} It is noteworthy that intradialytic hypotension can develop even among patients in whom no ultrafiltration is prescribed; the reason(s) for hypotension are not entirely clear, although many have attributed hemodynamic instability to the rapid exchange of solutes induced by high-flux, high-efficiency hemodialysis, with extra- to intracellular shifting of body water. Although there is a tendency to reduce the extracorporeal blood flow in patients prone to hypotension, there is little evidence that this provides any benefit. Reducing blood flow decreased the volume of the extracorporeal circuit in the past when parallel plate and coil dialyzers were used; however, there is little change in the volume of the extracorporeal circuit in response to changes in blood flow when hollow fiber dialyzers are used. Reducing blood flow may, however, result in reduction of the delivered dose of dialysis.

Continuous Renal Replacement Therapy

The continuous RRTs represent a spectrum of treatment modalities. In their initial description, the continuous therapies were provided using arteriovenous extracorporeal circuits.^{639–643} Although this approach provided technical simplicity, blood flow was dependent on the gradient between MAP and central venous pressure, and there was an increased risk of complications from prolonged arterial cannulation.⁶⁴⁴ As a result, the continuous arteriovenous therapies have largely been supplanted by pump-driven, venovenous CRRT.^{645–648} The modalities of venovenous CRRT vary, predominantly based on their mechanism of solute removal. With continuous venovenous hemofiltration (CVVH), solute transport occurs by convection; with continuous venovenous hemodialysis (CVVHD) by diffusion; and with continuous venovenous hemodiafiltration (CVVHDF) by a combination of the two.^{648–650} Although, at the same level of urea clearance, convective therapies provide enhanced clearance of higher molecular weight solutes compared with diffusive therapies, no clear clinical benefit has been demonstrated for CVVH or CVVHDF as compared with CVVHD.⁶⁵¹

The clearance of urea and other small solutes during CRRT is generally proportional to the total effluent flow rate (the sum of ultrafiltrate and dialysate flow rates),^{643,648,649} and the dose of therapy is usually expressed as the effluent volume indexed to body weight. This approach to estimating solute clearance is based on the assumption of near-complete solute equilibration between blood and effluent and may overestimate the actual solute clearance.^{652,653} Several single-center RCTs demonstrated an improvement in survival when doses of CVVH were increased from 20 to 25 mL/kg per hour to

doses in excess of 35 to 45 mL/kg per hour^{654,655}; however, other small studies did not find a similar benefit.^{584,656} Two large multicenter RCTs also did not find a survival benefit associated with more intensive CRRT.^{594,657} In the previously described ATN study, 1124 patients were randomized to two intensities of RRT.⁵⁹⁴ In both treatment arms, patients received IHD when hemodynamically stable and CVVHDF or SLED when hemodynamically unstable. In the less intensive arm, CVVHDF was provided at an effluent flow rate of 20 mL/kg per hour and, in the more intensive arm at 35 mL/kg per hour. All-cause mortality at 60 days was 51.5% in the less intensive arm and 53.6% in the more-intensive arm ($P = .47$; Fig. 29.5).⁵⁹⁴ In the Randomized Evaluation of Normal versus Augmented Level (RENAL) Replacement Therapy study, 1508 patients were randomized to CVVHDF at 25 mL/kg or 40 mL/kg per hour.⁶⁵⁷ All-cause mortality at 90 days was 44.7% in both treatment arms ($P = .99$; Fig. 29.6).⁶⁵⁷ Based on these data, the KDIGO Clinical Practice Guideline for Acute Kidney Injury recommends delivering an effluent volume during CRRT of 20 to 25 mL/kg per hour, recognizing that a slightly higher dose may need to be prescribed to achieve the target delivered dose to compensate for interruptions in treatment.³

Given the improved hemodynamic tolerance of CRRT as compared with IHD, particularly in patients with underlying hemodynamic instability, it has been postulated that CRRT would be associated with improved clinical outcomes. Five RCTs comparing outcomes with CRRT and IHD have been published. In a multicenter RCT of 166 patients with AKI, Mehta and colleagues observed ICU and hospital mortality rates of 59.5% and 65.5%, respectively, in patients randomized

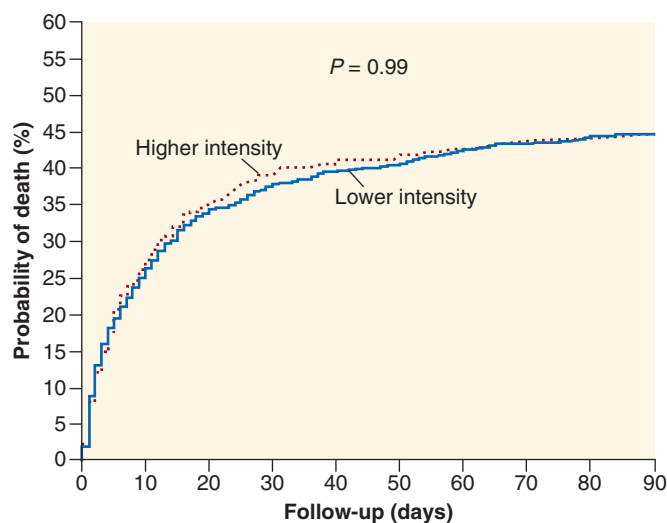


Fig. 29.6 90-Day mortality with intensive versus less intensive continuous venovenous hemodiafiltration (CVVHDF) in the Randomized Evaluation of Normal versus Augmented Level (RENAL) Replacement Therapy Study. Shown is the Kaplan-Meier plot of mortality in 1508 critically ill patients with acute kidney injury randomized to CVVHDF at 35 mL/kg vs. 20 mL/kg per hour. At 90 days, mortality was 44.7% in both treatment groups (odds ratio, 1.00; 95% confidence interval, 0.81–1.23; $P = .99$). (From RENAL Replacement Therapy Study Investigators; Bellomo R, Cass A, Cole L, et al. Intensity of continuous renal-replacement therapy in critically ill patients. *N Engl J Med*. 2009;361:1627–1638.)

to CRRT as compared with 41.5% and 47.6%, respectively, in patients randomized to IHD ($P < .02$).⁶⁵⁸ As the result of an imbalance in randomization, patients in the CRRT arm had a higher severity of illness as measured by the APACHE III score and a higher rate of liver failure. Adjusting for the imbalanced randomization in a post hoc analysis, the investigators found no difference in mortality attributable to modality of RRT. In another single-center randomized trial ($n = 80$), Augustine and colleagues reported more effective fluid removal and greater hemodynamic stability associated with CVVHD as compared with IHD, but no difference in survival.⁶⁵⁹ Similarly, in another single-center RCT from Switzerland, Uehlinger and colleagues observed no difference in survival in 70 patients randomized to CVVHDF as compared with 55 patients assigned to IHD.⁶⁶⁰ In the Hemodiafe study, a multicenter RCT conducted in 21 ICUs in France, Vinsonneau and colleagues reported 60-day survival rates of 31.5% in 184 patients randomized to IHD as compared with 32.6% in 175 patients randomized to CVVHDF ($P = .98$)⁶³² (Fig. 29.7). Similarly, Lins and colleagues observed hospital mortality rates of 62.5% in 144 patients randomized to IHD and 58.1% in 172 patients randomized to CRRT ($P = .43$).⁶⁶¹ Multiple meta-analyses have concluded that there is no difference in survival among patients undergoing RRT.^{662–664} Although several studies have suggested that CRRT is associated with improved rates of recovery of kidney function in surviving patients as compared with IHD,^{658,665–668} all these studies are confounded by higher mortality rates in the CRRT group. When analyzed across studies in which there were no differences in mortality, rates of recovery of kidney function did not appear to be affected by the modality of RRT.^{666,662,664,669}

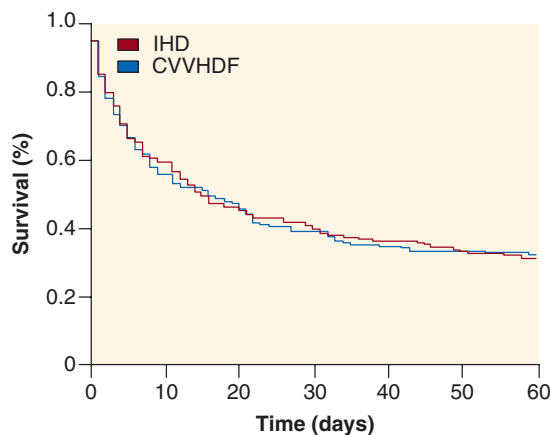


Fig. 29.7 60-Day survival with intermittent hemodialysis (IHD) versus continuous venovenous hemodiafiltration (CVVHDF) in the Hemodiafe Study. Shown is a Kaplan-Meier plot of survival among 359 critically ill patients with acute kidney injury randomized to intermittent hemodialysis versus continuous venovenous hemodiafiltration. At 60 days, survival was 31.5% among patients randomized to IHD versus 32.6% among patients randomized to CVVHDF ($P = .98$). (From Vinsonneau C, Camus C, Combes A, et al; Hemodiafe Study Group. Continuous venovenous haemodiafiltration versus intermittent haemodialysis for acute renal failure in patients with multiple-organ dysfunction syndrome: a multicentre randomised trial. *Lancet*. 2006; 368:379–385.)

Prolonged Intermittent Renal Replacement Therapy

Prolonged intermittent renal replacement therapy (PIRRT) represents a treatment modality in which conventional hemodialysis equipment is modified to provide extended-duration hemodialysis using lower blood flow rates and dialysate flow rates.^{670,671} A variety of terms have been developed to describe these therapies including SLED,^{672,673} extended daily dialysis (EDD),⁶⁷⁴ and sustained low-efficiency daily dialysis (SLEDD-f).⁶⁷⁵ By extending the duration of the dialysis treatment while providing slower ultrafiltration and solute clearance, these therapies are generally associated with enhanced hemodynamic tolerability compared with IHD. The degree of metabolic control attained with these treatments is comparable to that observed with CRRT.⁶⁷⁶ In an observational study performed in three ICUs in New Zealand, Australia, and Italy that changed from using CRRT to PIRRT, there was no difference in observed outcomes following the change in modality of RRT.⁶⁷¹ Similarly, in a single-center prospective RCT that included 232 patients, 90-day survival rates were similar in the PIRRT and CRRT groups (PIRRT, 50.4%; CRRT, 44.4%; $P = .43$), although overall resource use was lower with PIRRT.⁶⁷⁷ In a meta-analysis, there were no differences in mortality or recovery of kidney function comparing PIRRT with CRRT.⁶⁷⁸

Peritoneal Dialysis

The use of PD in the management of AKI has diminished as the use of continuous and hybrid therapies have increased.^{679–681} Peritoneal dialysis has the advantage of requiring minimal technology, facilitating its use in remote or resource-constrained areas.⁶⁸² As a result, it is still used in the treatment of AKI in regions where access to IHD or CRRT is not possible. Access for acute PD can be obtained by percutaneous placement of an uncuffed temporary peritoneal catheter or through surgical placement of a tunneled cuffed catheter. PD has the advantage of avoiding the need for vascular access or anticoagulation. Solute clearance and control of metabolic parameters may be inferior to that achieved with other modalities of RRT.⁶⁸³ Although systemic hypotension is less of an issue than with other modalities of RRT, ultrafiltration cannot be as tightly controlled. Other limitations include the relative contraindication in patients with acute abdominal processes or recent abdominal surgery, the risk of visceral organ injury during catheter placement, the risk of PD-associated peritonitis, and an increased tendency toward hyperglycemia due to the high glucose concentrations in peritoneal dialysate, which in other acute settings has been associated with adverse outcomes.

Several trials have compared outcomes using PD with other modalities of RRT in AKI.^{683–686} In a study of 70 patients with infection-associated AKI in Vietnam, 58 of whom had severe falciparum malaria, PD was associated with less adequate metabolic control and higher mortality than continuous hemofiltration.⁶⁸³ In contrast, in a study of 120 patients in Brazil who were randomized to high-volume PD or daily hemodialysis, indices of metabolic control, recovery of kidney function, and survival were similar with both modalities.⁶⁸⁴ In a meta-analysis that included eight observational studies and four clinical trials, Chionh and colleagues observed similar survival rates with peritoneal dialysis as compared with extracorporeal RRT in patients with AKI.⁶⁸⁷

SUMMARY OF ACUTE KIDNEY INJURY MANAGEMENT

Acute kidney injury remains a common and serious disease with protean causes and variable clinical courses. The management of AKI begins with prevention in those clinical circumstances in which evidence-based preventive interventions are available. Although pharmacologic therapy is available for select causes of AKI, the treatment of established ATN is largely supportive, with pharmacology and RRTs designed to ameliorate the adverse metabolic and clinical complications of this condition. Additional research is needed to identify treatments that decrease the risk of developing AKI and, in patients with established AKI, reduce the severity of AKI and/or facilitate recovery.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

- Which of the following is true regarding the role of early goal-directed therapy (EGDT) for the management of sepsis?
 - EGDT is associated with a reduction in dialysis-requiring acute kidney injury (AKI) but not mortality.
 - EGDT is associated with a reduction in dialysis-requiring AKI and mortality.
 - Recent clinical trials have demonstrated that EGDT is not associated with a reduction in dialysis-requiring AKI.
 - Early antibiotic therapy and volume resuscitation are not associated with improved patient outcomes.

Answer: c

Rationale: The recent ProCESS, ARISE, and ProMISE trials did not demonstrate a reduction in the need for renal support with EGDT.

- A 63-year-old woman with diabetes mellitus complicated by retinopathy and nephropathy is admitted with a non-ST-elevation myocardial infarction. Her serum creatinine level is 1.68 mg/dL and her urine albumin excretion is 762 μ g/mg creatinine. Which one of the following interventions is associated with the strongest evidence for reducing her risk of developing contrast-associated acute kidney injury following her planned coronary angiography?
 - Oral N-acetylcysteine
 - Periprocedural hemofiltration
 - Periprocedural infusion of isotonic sodium chloride
 - Use of iodixanol

Answer: c

Rationale: Periprocedural infusion of isotonic sodium chloride is the strongest evidence based on the results of the PRESERVE trial, absence of sound evidence of any benefit with hemofiltration, and studies demonstrating no difference between iodixanol and low osmolal contrast agents with regard to risk for AKI.

- Which one of the following strategies for the management of renal replacement therapy in patients with acute kidney

injury and multisystem organ failure in the setting of sepsis is most strongly supported by evidence from clinical trials?

- Use of continuous renal replacement therapy rather than intermittent hemodialysis
- Use of continuous hemofiltration rather than continuous hemodialysis
- Use of lower doses (20–25 mL/kg per hour) rather than higher-doses (>35 mL/kg per hour) of continuous renal replacement therapy
- Use of daily intermittent hemodialysis with a delivered Kt/V_{urea} of 1.2 to 1.4 per treatment rather than alternate-day hemodialysis with the same delivered Kt/V_{urea} per treatment

Answer: c

Rationale: Use of lower doses (20–25 mL/kg per hour) rather than higher-doses (>35 mL/kg per hour) of continuous renal replacement therapy is the correct option based on the results of the ATN and RENAL trials and consistent with the KDIGO AKI Clinical Practice Guidelines.

- With regard to the optimal time to initiate renal replacement therapy in the setting of acute kidney injury, which of the following is true?
 - No clinical trials have compared early versus late initiation of renal replacement therapy for acute kidney injury.
 - Early initiation of renal replacement therapy is associated with improved recovery of kidney function and lower mortality.
 - The costs of early initiation of renal replacement therapy outweigh the potential clinical benefit.
 - There are currently insufficient data to support the clinical benefit of early initiation of renal replacement therapy.

Answer: d

Rationale: Trials to date have generated conflicting data on the benefit of early initiation of renal replacement therapy.